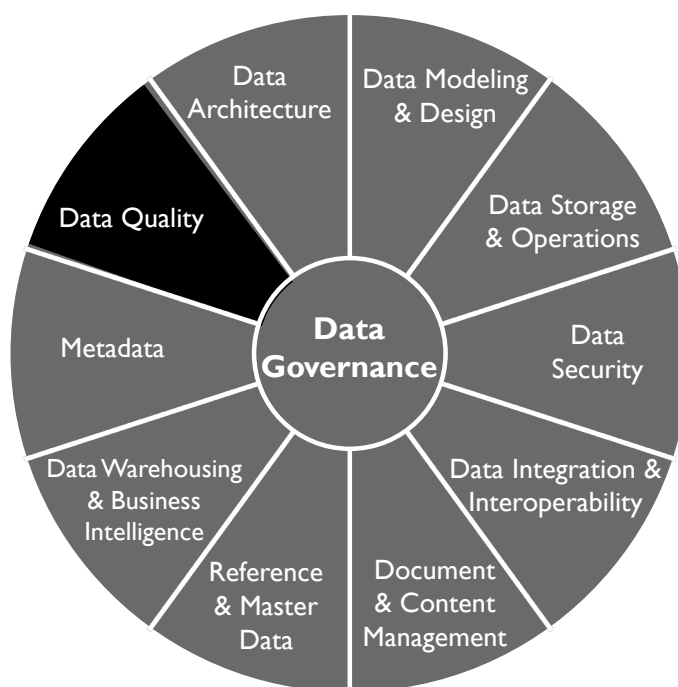


Data Quality



DAMA-DMBOK2 Data Management Framework

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1. Introduction

Effective data management involves a set of complex, interrelated processes that enable an organization to use its data to achieve strategic goals. Data management includes the ability to design data for applications, store and access it securely, share it appropriately, learn from it, and ensure it meets business needs. An assumption underlying assertions about the value of data is that the data itself is reliable and trustworthy. In other words, that it is of high quality.

However, many factors can undermine that assumption by contributing to poor quality data: Lack of understanding about the effects of poor quality data on organizational success, bad planning, ‘siloe’ system design, inconsistent development processes, incomplete documentation, a lack of standards, or a lack of governance. Many organizations fail to define what makes data fit for purpose.

All data management disciplines contribute to the quality of data, and high quality data that supports the organization should be the goal of all data management disciplines. Because uninformed decisions or actions by anyone who interacts with data can result in poor quality data, producing high quality data requires cross-functional commitment and coordination. Organizations and teams should be aware of this and should plan for high quality data, by executing processes and projects in ways that account for risk related to unexpected or unacceptable conditions in the data.

Because no organization has perfect business processes, perfect technical processes, or perfect data management practices, all organizations experience problems related to the quality of their data. Organizations that formally manage the quality of data have fewer problems than those that leave data quality to chance.

Formal data quality management is similar to continuous quality management for other products. It includes managing data through its lifecycle by setting standards, building quality into the processes that create, transform, and store data, and measuring data against standards. Managing data to this level usually requires a Data Quality program team. The Data Quality program team is responsible for engaging both business and technical data management professionals and driving the work of applying quality management techniques to data to ensure that data is fit for consumption for a variety of purposes. The team will likely be involved with a series of projects through which they can establish processes and best practices while addressing high priority data issues.

Because managing the quality of data involves managing the data lifecycle, a Data Quality program will also have operational responsibilities related to data usage. For example, reporting on data quality levels and engaging in the analysis, quantification, and prioritization of data issues. The team is also responsible for working with those who need data to do their jobs to ensure the data meets their needs and working with those who create, update, or delete data in the course of their jobs to ensure they are properly handling the data. Data quality depends on all who interact with the data, not just data management professionals.

As is the case with Data Governance and with data management as a whole, Data Quality Management is a program, not a project. It will include both project and maintenance work, along with a commitment to communications and training. Most importantly, the long-term success of data quality improvement program depends on getting an organization to change its culture and adopt a quality mindset. As stated in *The Leader’s Data Manifesto*: fundamental, lasting change requires committed leadership and involvement from people at all levels in an organization. People who use data to do their jobs – which in most organizations is a very large percentage of employees – need to drive change. One of the most critical changes to focus on is how their organizations manage and improve the quality of their data.⁷¹

⁷¹ For the full text of *The Leader’s Data Manifesto*, see <http://bit.ly/2sQhcy7>.

Data Quality Management

Definition: The planning, implementation, and control of activities that apply quality management techniques to data, in order to assure it is fit for consumption and meets the needs of data consumers.

Goals:

1. Develop a governed approach to make data fit for purpose based on data consumers' requirements.
2. Define standards, requirements, and specifications for data quality controls as part of the data lifecycle.
3. Define and implement processes to measure, monitor, and report on data quality levels.
4. Identify and advocate for opportunities to improve the quality of data, through process and system improvements.



Figure 91 Context Diagram: Data Quality

1.1 Business Drivers

The business drivers for establishing a formal Data Quality Management program include:

- Increasing the value of organizational data and the opportunities to use it
- Reducing risks and costs associated with poor quality data
- Improving organizational efficiency and productivity
- Protecting and enhancing the organization's reputation

Organizations that want to get value out of their data recognize that high quality data is more valuable than low quality data. Poor quality data is risk-laden (see Chapter 1). It can damage an organization's reputation, resulting in fines, lost revenue, lost customers, and negative media exposure. Regulatory requirements often demand high quality data. In addition, many direct costs are associated with poor quality data. For example,

- Inability to invoice correctly
- Increased customer service calls and decreased ability to resolve them
- Revenue loss due to missed business opportunities
- Delay of integration during mergers and acquisitions
- Increased exposure to fraud
- Loss due to bad business decisions driven by bad data
- Loss of business due to lack of good credit standing

Still high quality data is not an end in itself. It is a means to organizational success. Trustworthy data not only mitigates risk and reduces costs, but also improves efficiency. Employees can answer questions more quickly and consistently, when they are working with reliable data. They spend less time trying to figure out if the data is right and more time using the data to gain insight, make decisions, and serve customers.

1.2 Goals and Principles

Data Quality programs focus on these general goals:

- Developing a governed approach to make data fit for purpose based on data consumers' requirements
- Defining standards and specifications for data quality controls as part of the data lifecycle
- Defining and implementing processes to measure, monitor, and report on data quality levels
- Identifying and advocating for opportunities to improve the quality of data, through changes to processes and systems and engaging in activities that measurably improve the quality of data based on data consumer requirements

Data Quality programs should be guided by the following principles:

- **Criticality:** A Data Quality program should focus on the data most critical to the enterprise and its customers. Priorities for improvement should be based on the criticality of the data and on the level of risk if data is not correct.

- **Lifecycle management:** The quality of data should be managed across the data lifecycle, from creation or procurement through disposal. This includes managing data as it moves within and between systems (i.e., each link in the data chain should ensure data output is of high quality).
- **Prevention:** The focus of a Data Quality program should be on preventing data errors and conditions that reduce the usability of data; it should not be focused on simply correcting records.
- **Root cause remediation:** Improving the quality of data goes beyond correcting errors. Problems with the quality of data should be understood and addressed at their root causes, rather than just their symptoms. Because these causes are often related to process or system design, improving data quality often requires changes to processes and the systems that support them.
- **Governance:** Data Governance activities must support the development of high quality data and Data Quality program activities must support and sustain a governed data environment.
- **Standards-driven:** All stakeholders in the data lifecycle have data quality requirements. To the degree possible, these requirements should be defined in the form of measurable standards and expectations against which the quality of data can be measured.
- **Objective measurement and transparency:** Data quality levels need to be measured objectively and consistently. Measurements and measurement methodology should be shared with stakeholders since they are the arbiters of quality.
- **Embedded in business processes:** Business process owners are responsible for the quality of data produced through their processes. They must enforce data quality standards in their processes.
- **Systematically enforced:** System owners must systematically enforce data quality requirements.
- **Connected to service levels:** Data quality reporting and issues management should be incorporated into Service Level Agreements (SLA).

1.3 Essential Concepts

1.3.1 Data Quality

The term *data quality* refers both to the characteristics associated with high quality data and to the processes used to measure or improve the quality of data. These dual usages can be confusing, so it helps to separate them and clarify what constitutes high quality data.⁷²

⁷² In the DAMA-DMBOK2, we have tried to avoid using the words data quality without clarifying their context. For example, referring to *high quality data* or *low quality data*, and to *data quality work efforts* or *data quality activities*.

Data is of high quality to the degree that it meets the expectations and needs of data consumers. That is, if the data is fit for the purposes to which they want to apply it. It is of low quality if it is not fit for those purposes. Data quality is thus dependent on context and on the needs of the data consumer.

One of the challenges in managing the quality of data is that expectations related to quality are not always known. Customers may not articulate them. Often, the people managing data do not even ask about these requirements. However, if data is to be reliable and trustworthy, then data management professionals need to better understand their customers' quality requirements and how to measure them. This needs to be an ongoing discussion, as requirements change over time as business needs and external forces evolve.

1.3.2 Critical Data

Most organizations have a lot of data, not all of which is of equal importance. One principle of Data Quality Management is to focus improvement efforts on data that is most important to the organization and its customers. Doing so gives the program scope and focus and enables it to make a direct, measurable impact on business needs.

While specific drivers for criticality will differ by industry, there are common characteristics across organizations. Data can be assessed based on whether it is required by:

- Regulatory reporting
- Financial reporting
- Business policy
- Ongoing operations
- Business strategy, especially efforts at competitive differentiation

Master Data is critical by definition. Data sets or individual data elements can be assessed for criticality based on the processes that consume them, the nature of the reports they appear in, or the financial, regulatory, or reputational risk to the organization if something were to go wrong with the data.⁷³

1.3.3 Data Quality Dimensions

A *Data Quality dimension* is a measurable feature or characteristic of data. The term *dimension* is used to make the connection to dimensions in the measurement of physical objects (e.g., length, width, height). Data quality dimensions provide a vocabulary for defining data quality requirements. From there, they can be used to define results of initial data quality assessment as well as ongoing measurement. In order to measure the quality of data, an organization needs to establish characteristics that are both important to business processes (worth measuring) and measurable. Dimensions provide a basis for measurable rules, which themselves should be directly connected to potential risks in critical processes.

⁷³ See Jugulum (2014), Chapters 6 and 7 for an approach to rationalizing critical data.

For example, if the data in the customer email address field is incomplete, then we will not be able to send product information to our customers via email, and we will lose potential sales. Therefore, we will measure the percentage of customers for whom we have usable email addresses, and we will improve our processes until we have a usable email address for at least 98% of our customers.

Many leading thinkers in data quality have published sets of dimensions.⁷⁴ The three most influential are described here because they provide insight into how to think about what it means to have high quality data, as well as into how data quality can be measured.

The Strong-Wang framework (1996) focuses on data consumers' perceptions of data. It describes 15 dimensions across four general categories of data quality:

- Intrinsic DQ
 - Accuracy
 - Objectivity
 - Believability
 - Reputation
- Contextual DQ
 - Value-added
 - Relevancy
 - Timeliness
 - Completeness
 - Appropriate amount of data
- Representational DQ
 - Interpretability
 - Ease of understanding
 - Representational consistency
 - Concise representation
- Accessibility DQ
 - Accessibility
 - Access security

In *Data Quality for the Information Age* (1996), Thomas Redman formulated a set of data quality dimension rooted in data structure.⁷⁵ Redman defines a data item as a “representable triple”: a value from the domain of an attribute within an entity. Dimensions can be associated with any of the component pieces of data – the model (entities and attributes) as well as the values. Redman includes the dimension of representation, which he defines as a set of rules for recording data items. Within these three general categories (data model, data values, representation), he describes more than two dozen dimensions. They include the following:

⁷⁴ In addition to the examples detailed here and numerous academic papers on this topic, see Loshin (2001), Olson (2003), McGilvray (2008), and Sebastian-Coleman (2013) for detailed discussions on data quality dimensions. See Myers (2013) for a comparison of dimensions.

⁷⁵ Redman expanded and revised his set of dimensions in *Data Quality: The Field Guide* (2001).

Data Model:

- Content:
 - Relevance of data
 - The ability to obtain the values
 - Clarity of definitions
- Level of detail:
 - Attribute granularity
 - Precision of attribute domains
- Composition:
 - Naturalness: The idea that each attribute should have a simple counterpart in the real world and that each attribute should bear on a single fact about the entity
 - Identify-ability: Each entity should be distinguishable from every other entity
 - Homogeneity
 - Minimum necessary redundancy
- Consistency:
 - Semantic consistency of the components of the model
 - Structure consistency of attributes across entity types
- Reaction to change:
 - Robustness
 - Flexibility

Data Values:

- Accuracy
- Completeness
- Currency
- Consistency

Representation:

- Appropriateness
- Interpretability
- Portability
- Format precision
- Format flexibility
- Ability to represent null values
- Efficient use of storage
- Physical instances of data being in accord with their formats

Redman recognizes that consistency of entities, values, and representation can be understood in terms of constraints. Different types of consistency are subject to different kinds of constraints.

In *Improving Data Warehouse and Business Information Quality* (1999), Larry English presents a comprehensive set of dimensions divided into two broad categories: inherent and pragmatic.⁷⁶ Inherent characteristics are independent of data use. Pragmatic characteristics are associated with data presentation and are dynamic; their value (quality) can change depending on the uses of data.

- **Inherent** quality characteristics
 - Definitional conformance
 - Completeness of values
 - Validity or business rule conformance
 - Accuracy to a surrogate source
 - Accuracy to reality
 - Precision
 - Non-duplication
 - Equivalence of redundant or distributed data
 - Concurrency of redundant or distributed data
- **Pragmatic** quality characteristics
 - Accessibility
 - Timeliness
 - Contextual clarity
 - Usability
 - Derivation integrity
 - Rightness or fact completeness

In 2013, DAMA UK produced a white paper describing six core dimensions of data quality:

- **Completeness:** The proportion of data stored against the potential for 100%.
- **Uniqueness:** No entity instance (thing) will be recorded more than once based upon how that thing is identified.
- **Timeliness:** The degree to which data represent reality from the required point in time.
- **Validity:** Data is valid if it conforms to the syntax (format, type, range) of its definition.
- **Accuracy:** The degree to which data correctly describes the ‘real world’ object or event being described.
- **Consistency:** The absence of difference, when comparing two or more representations of a thing against a definition.

The DAMA UK white paper also describes other characteristics that have an impact on quality. While the white paper does not call these dimensions, they work in a manner similar to Strong and Wang’s contextual and representational DQ and English’s pragmatic characteristics.

- **Usability:** Is the data understandable, simple, relevant, accessible, maintainable and at the right level of precision?

⁷⁶ English expanded and revised his dimensions in *Information Quality Applied* (2009).

- **Timing issues** (beyond timeliness itself): Is it stable yet responsive to legitimate change requests?
- **Flexibility**: Is the data comparable and compatible with other data? Does it have useful groupings and classifications? Can it be repurposed? Is it easy to manipulate?
- **Confidence**: Are Data Governance, Data Protection, and Data Security processes in place? What is the reputation of the data, and is it verified or verifiable?
- **Value**: Is there a good cost / benefit case for the data? Is it being optimally used? Does it endanger people's safety or privacy, or the legal responsibilities of the enterprise? Does it support or contradict the corporate image or the corporate message?

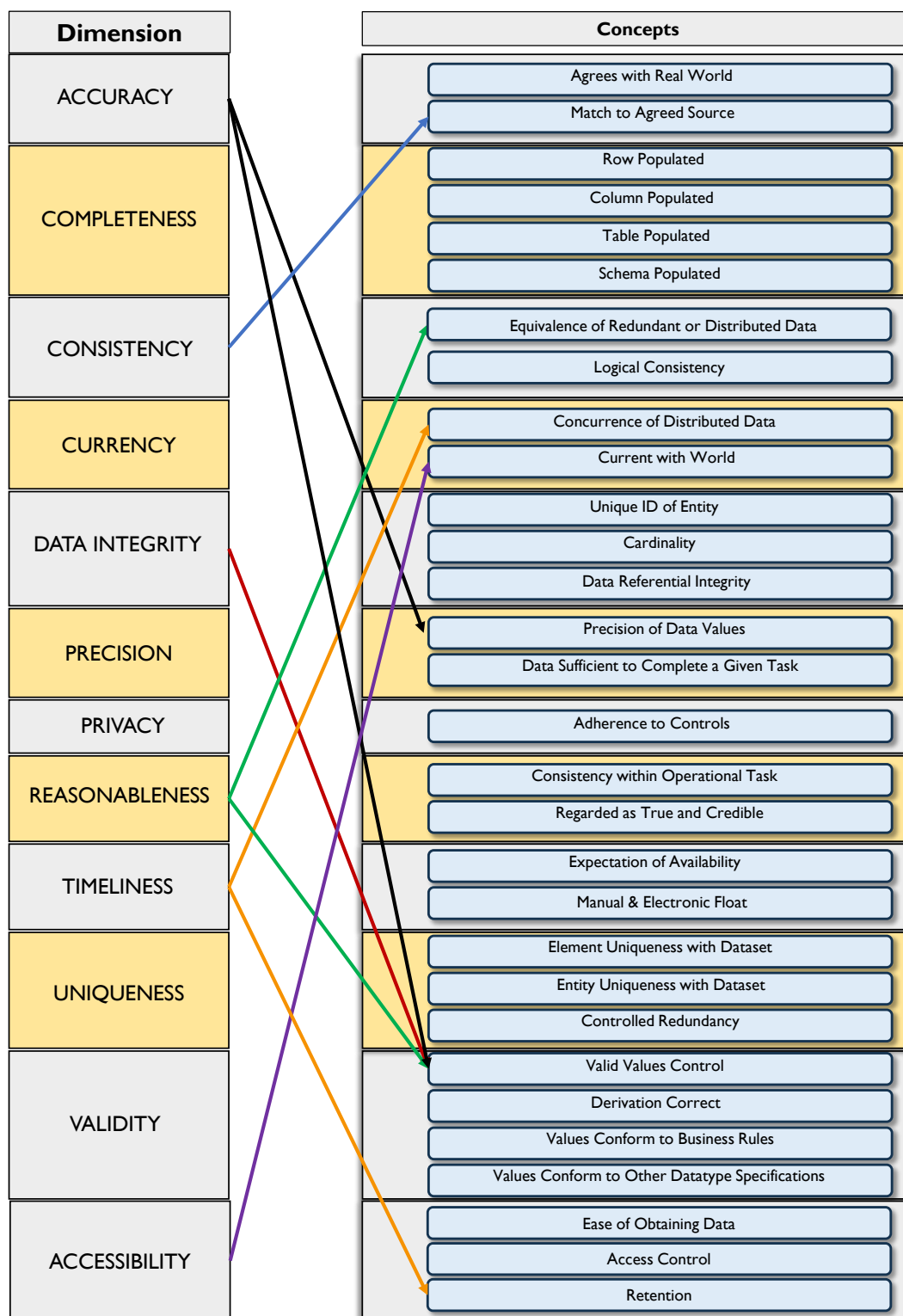
While there is not a single, agreed-to set of data quality dimensions, these formulations contain common ideas. Dimensions include some characteristics that can be measured objectively (completeness, validity, format conformity) and others that depend on heavily context or on subjective interpretation (usability, reliability, reputation). Whatever names are used, dimensions focus on whether there is enough data (completeness), whether it is right (accuracy, validity), how well it fits together (consistency, integrity, uniqueness), whether it is up-to-date (timeliness), accessible, usable, and secure. Table 29 contains definitions of a set of data quality dimensions, about which there is general agreement and describes approaches to measuring them.

Table 29 Common Dimensions of Data Quality

Dimension of Quality	Description
Accuracy	Accuracy refers to the degree that data correctly represents 'real-life' entities. Accuracy is difficult to measure, unless an organization can reproduce data collection or manually confirm accuracy of records. Most measures of accuracy rely on comparison to a data source that has been verified as accurate, such as a system of record or data from a reliable source (e.g., Dun and Bradstreet Reference Data).
Completeness	Completeness refers to whether all required data is present. Completeness can be measured at the data set, record, or column level. Does the data set contain all the records expected? Are records populated correctly? (Records with different statuses may have different expectations for completeness.) Are columns/attributes populated to the level expected? (Some columns are mandatory. Optional columns are populated only under specific conditions.) Assign completeness rules to a data set with varying levels of constraint: Mandatory attributes that require a value, data elements with conditional and optional values, and inapplicable attribute values. Data set level measurements may require comparison to a source of record or may be based on historical levels of population.
Consistency	Consistency can refer to ensuring that data values are consistently represented within a data set and between data sets, and consistently associated across data sets. It can also refer to the size and composition of data sets between systems or across time. Consistency may be defined between one set of attribute values and another attribute set within the same record (record-level consistency), between one set of attribute values and another attribute set in different records (cross-record consistency), or between one set of attribute values and the same attribute set within the same record at different points in time (temporal consistency). Consistency can also be used to refer to consistency of format. Take care not to confuse consistency with accuracy or correctness. Characteristics that are expected to be consistent within and across data sets can be used as the basis for standardizing data. Data Standardization refers to the conditioning of input data to ensure that data meets rules for content and format. Standardizing data enables more effective matching and facilitates consistent output. Encapsulate consistency constraints as a set of rules that specify consistent relationships between values of attributes, either across a record or message, or along all values of a single attribute (such as a range or list of valid values). For example, one might expect that the number of transactions each day does not exceed 105% of the running average number of transactions for the previous 30 days.

Dimension of Quality	Description
Integrity	Data Integrity (or Coherence) includes ideas associated with completeness, accuracy, and consistency. In data, integrity usually refers to either referential integrity (consistency between data objects via a reference key contained in both objects) or internal consistency within a data set such that there are no holes or missing parts. Data sets without integrity are seen as corrupted, or have data loss. Data sets without <i>referential</i> integrity have ‘orphans’ – invalid reference keys, or ‘duplicates’ – identical rows which may negatively affect aggregation functions. The level of orphan records can be measured as a raw count or as a percentage of the data set.
Reasonability	Reasonability asks whether a data pattern meets expectations. For example, whether a distribution of sales across a geographic area makes sense based on what is known about the customers in that area. Measurement of reasonability can take different forms. For example, reasonability may be based on comparison to benchmark data, or past instances of a similar data set (e.g., sales from the previous quarter). Some ideas about reasonability may be perceived as subjective. If this is the case, work with data consumers to articulate the basis of their expectations of data to formulate objective comparisons. Once benchmark measurements of reasonability are established, these can be used to objectively compare new instances of the same data set in order to detect change. (See Section 4.5.)
Timeliness	The concept of data Timeliness refers to several characteristics of data. Measures of timeliness need to be understood in terms of expected volatility – how frequently data is likely to change and for what reasons. Data currency is the measure of whether data values are the most up-to-date version of the information. Relatively static data, for example some Reference Data values like country codes, may remain current for a long period. Volatile data remains current for a short period. Some data, for example, stock prices on financial web pages, will often be shown with an as-of-time, so that data consumers understand the risk that the data has changed since it was recorded. During the day, while the markets are open, such data will be updated frequently. Once markets close, the data will remain unchanged, but will still be current, since the market itself is inactive. Latency measures the time between when the data was created and when it was made available for use. For example, overnight batch processing can give a latency of 1 day at 8am for data entered into the system during the prior day, but only one hour for data generated during the batch processing. (See Chapter 8.)
Uniqueness / Deduplication	Uniqueness states that no entity exists more than once within the data set. Asserting uniqueness of the entities within a data set implies that a key value relates to each unique entity, and only that specific entity, within the data set. Measure uniqueness by testing against key structure. (See Chapter 5.)
Validity	Validity refers to whether data values are consistent with a defined domain of values. A domain of values may be a defined set of valid values (such as in a reference table), a range of values, or value that can be determined via rules. The data type, format, and precision of expected values must be accounted for in defining the domain. Data may also only be valid for a specific length of time, for example data that is generated from RFID (radio frequency ID) or some scientific data sets. Validate data by comparing it to domain constraints. Keep in mind that data may be valid (i.e., it may meet domain requirements) and still not be accurate or correctly associated with particular records.

Figure 92 aligns data quality dimensions and concepts associated with those dimensions. The arrows indicate significant overlaps between concepts and also demonstrate that there is not agreement on a specific set. For example, the dimension of accuracy is associated with ‘agrees with real world’ and ‘match to agreed source’ and also to the concepts associated with validity, such as ‘derivation correct’.

Figure 92 Relationship Between Data Quality Dimensions⁷⁷⁷⁷ Adapted from Myers (2013), used with permission.

1.3.4 Data Quality and Metadata

Metadata is critical to managing the quality of data. The quality of data is based on how well it meets the requirements of data consumers. Metadata defines what the data represents. Having a robust process by which data is defined supports the ability of an organization to formalize and document the standards and requirements by which the quality of data can be measured. Data quality is about meeting expectations. Metadata is a primary means of clarifying expectations.

Well-managed Metadata can also support the effort to improve the quality of data. A Metadata repository can house results of data quality measurements so that these are shared across the organization and the Data Quality team can work toward consensus about priorities and drivers for improvement. (See Chapter 12.)

1.3.5 Data Quality ISO Standard

ISO 8000, the international standard for data quality, is being developed to enable the exchange of complex data in an application-neutral form. In the introduction to the standard, ISO asserts: “The ability to create, collect, store, maintain, transfer, process and present data to support business processes in a timely and cost effective manner requires both an understanding of the characteristics of the data that determine its quality, and an ability to measure, manage and report on data quality.”

ISO 8000 defines characteristics that can be tested by any organization in the data supply chain to objectively determine conformance of the data to ISO 8000.⁷⁸

The first published part of ISO 8000 (part 110, published in 2008) focused on the syntax, semantic encoding, and conformance to the data specification of Master Data. Other parts projected for the standard include part 100 - Introduction, part 120 - Provenance, part 130 -Accuracy, and part 140 - Completeness.⁷⁹

ISO defines quality data as “portable data that meets stated requirements.”⁸⁰ The data quality standard is related to the ISO’s overall work on data portability and preservation. Data is considered ‘portable’ if it can be separated from a software application. Data that can only be used or read using a specific licensed software application is subject to the terms of the software license. An organization may not be able to use data it created unless that data can be detached from the software that was used to create it.

To meet stated requirements requires that these requirements be defined in a clear, unambiguous manner. ISO 8000 is supported through ISO 22745, a standard for defining and exchanging Master Data. ISO 22745 defines how data requirement statements should be constructed, provides examples in XML, and defines a format for

⁷⁸ <http://bit.ly/2ttdiZJ>.

⁷⁹ <http://bit.ly/2sANGdi>.

⁸⁰ <http://bit.ly/2rV1oWC>.

the exchange of encoded data.⁸¹ ISO 22745 creates portable data by labeling the data using an ISO 22745 compliant Open Technical Dictionary such as the ECCMA Open Technical Dictionary (eOTD).

The intention of ISO 8000 is to help organizations define what is and is not quality data, enable them to ask for quality data using standard conventions, and verify that they have received quality data using those same standards. When standards are followed, requirements can be confirmed through a computer program.

ISO 8000 - Part 61 Information and data quality management process reference model is under development.⁸² This standard will describe the structure and organization of data quality management, including:

- Data Quality Planning
- Data Quality Control
- Data Quality Assurance
- Data Quality Improvement

1.3.6 Data Quality Improvement Lifecycle

Most approaches to improving data quality are based on the techniques of quality improvement in the manufacture of physical products.⁸³ In this paradigm, data is understood as the product of a set of processes. At its simplest, a process is defined as a series of steps that turns inputs into outputs. A process that creates data may consist of one-step (data collection) or many steps: data collection, integration into a data warehouse, aggregation in a data mart, etc. At any step, data can be negatively affected. It can be collected incorrectly, dropped or duplicated between systems, aligned or aggregated incorrectly, etc. Improving data quality requires the ability to assess the relationship between inputs and outputs, in order to ensure that inputs meet the requirements of the process and that outputs conform to expectations. Since outputs from one process become inputs to other processes, requirements must be defined along the whole data chain.

A general approach to data quality improvement, shown in Figure 93, is a version of the Shewhart / Deming cycle.⁸⁴ Based on the scientific method, the Shewhart / Deming cycle is a problem-solving model known as ‘plan-do-check-act’. Improvement comes through a defined set of steps. The condition of data must be measured against standards and, if it does not meet standards, root cause(s) of the discrepancy from standards must be identified and remediated. Root causes may be found in any of the steps of the process, technical or non-technical. Once remediated, data should be monitored to ensure that it continues to meet requirements.

⁸¹ <http://bit.ly/2rUZyoz>.

⁸² <http://bit.ly/2sVik3Q>.

⁸³ See Wang (1998), English (1999), Redman (2001), Loshin (2001), and McGilvray (2008). See Pierce (2004) for an overview of literature related to the concept of data as a product.

⁸⁴ See American Society for Quality: <http://bit.ly/1lelyBK> Plan-Do-Check-Act was originated by Walter Shewhart and popularized by W. Edwards Deming. 6 Sigma’s Measure, Analyze, Improve, Control (DMAIC) is a variation on this cycle.

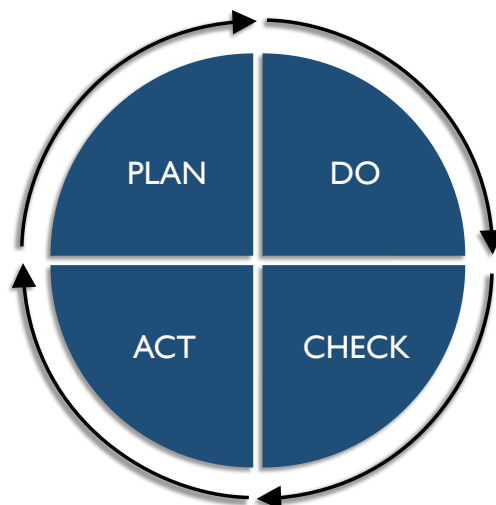


Figure 93 The Shewhart Chart

For a given data set, a Data Quality Management cycle begins by identifying the data that does not meet data consumers' requirements and data issues that are obstacles to the achievement of business objectives. Data needs to be assessed against key dimensions of quality and known business requirements. Root causes of issues will need to be identified so that stakeholders can understand the costs of remediation and the risks of not remediating the issues. This work is often done in conjunction with Data Stewards and other stakeholders.

In the *Plan* stage, the Data Quality team assesses the scope, impact, and priority of known issues, and evaluates alternatives to address them. This plan should be based on a solid foundation of analysis of the root causes of issues. From knowledge of the causes and the impact of the issues, cost / benefit can be understood, priority can be determined, and a basic plan can be formulated to address them.

In the *Do* stage, the DQ team leads efforts to address the root causes of issues and plan for ongoing monitoring of data. For root causes that are based on non-technical processes, the DQ team can work with process owners to implement changes. For root causes that require technical changes, the DQ team should work with technical teams to ensure that requirements are implemented correctly and that technical changes do not introduce errors.

The *Check* stage involves actively monitoring the quality of data as measured against requirements. As long as data meets defined thresholds for quality, additional actions are not required. The processes will be considered under control and meeting business requirements. However, if the data falls below acceptable quality thresholds, then additional action must be taken to bring it up to acceptable levels.

The *Act* stage is for activities to address and resolve emerging data quality issues. The cycle restarts, as the causes of issues are assessed and solutions proposed. Continuous improvement is achieved by starting a new cycle. New cycles begin as:

- Existing measurements fall below thresholds
- New data sets come under investigation
- New data quality requirements emerge for existing data sets
- Business rules, standards, or expectations change

The cost of getting data right the first time is cheaper than the costs from getting data wrong and fixing it later. Building quality into the data management processes from the beginning costs less than retrofitting it. Maintaining high quality data throughout the data lifecycle is less risky than trying to improve quality in an existing process. It also creates a far lower impact on the organization. Establishing criteria for data quality at the beginning of a process or system build is one sign of a mature Data Management Organization. Doing so takes governance and discipline, as well as cross-functional collaboration.

1.3.7 Data Quality Business Rule Types

Business rules describe how business should operate internally, in order to be successful and compliant with the outside world. Data Quality Business Rules describe how data should exist in order to be useful and usable within an organization. These rules can be aligned with dimensions of quality and used to describe data quality requirements. For example, a business rule that all state code fields must comply with the US State Abbreviations can be enforced by data entry pick lists and data integration lookups. The level of valid or invalid records can then be measured.

Business rules are commonly implemented in software, or by using document templates for data entry. Some common simple business rule types are:

- **Definitional conformance:** Confirm that the same understanding of data definitions is implemented and used properly in processes across the organization. Confirmation includes algorithmic agreement on calculated fields, including any time, or local constraints, and rollup and status interdependence rules.
- **Value presence and record completeness:** Rules defining the conditions under which missing values are acceptable or unacceptable.
- **Format compliance:** One or more patterns specify values assigned to a data element, such as standards for formatting telephone numbers.
- **Value domain membership:** Specify that a data element's assigned value is included in those enumerated in a defined data value domain, such as 2-Character United States Postal Codes for a STATE field.
- **Range conformance:** A data element assigned value must be within a defined numeric, lexicographic, or time range, such as greater than 0 and less than 100 for a numeric range.
- **Mapping conformance:** Indicating that the value assigned to a data element must correspond to one selected from a value domain that maps to other equivalent corresponding value domain(s). The STATE data domain again provides a good example, since State values may be represented using different value domains (USPS Postal codes, FIPS 2-digit codes, full names), and these types of rules validate that 'AL' and '01' both map to 'Alabama.'

- **Consistency rules:** Conditional assertions that refer to maintaining a relationship between two (or more) attributes based on the actual values of those attributes. For example, address validation where postal codes correspond to particular States or Provinces.
- **Accuracy verification:** Compare a data value against a corresponding value in a system of record or other verified source (e.g., marketing data purchased from a vendor) to verify that the values match.
- **Uniqueness verification:** Rules that specify which entities must have a unique representation and whether one and only one record exists for each represented real world object.
- **Timeliness validation:** Rules that indicate the characteristics associated with expectations for accessibility and availability of data.

Other types of rules may involve aggregating functions applied to sets of data instances (see Section 4.5). Examples of aggregation checks include:

- Validate reasonableness of the number of records in a file. This requires keeping statistics over time to generate trends.
- Validate reasonableness of an average amount calculated from a set of transactions. This requires establishing thresholds for comparison, and may be based on statistics over time.
- Validate the expected variance in the count of transactions over a specified timeframe. This requires keeping statistics over time and using them to establish thresholds.

1.3.8 Common Causes of Data Quality Issues

Data quality issues can emerge at any point in the data lifecycle, from creation to disposal. When investigating root causes, analysts should look for potential culprits, like problems with data entry, data processing, system design, and manual intervention in automated processes. Many issues will have multiple causes and contributing factors (especially if people have created ways to work around them). These causes of issues also imply ways to prevent issues: through improvement to interface design, testing of data quality rules as part of processing, a focus on data quality within system design, and strict controls on manual intervention in automated processes.

1.3.8.1 Issues Caused by Lack of Leadership

Many people assume that most data quality issues are caused by data entry errors. A more sophisticated understanding recognizes that gaps in or poor execution of business and technical processes cause many more problems than mis-keying. However, common sense says and research indicates that many data quality problems are caused by a lack of organizational commitment to high quality data, which itself stems from a lack of leadership, in the form of both governance and management.

Every organization has information and data assets that are of value to its operations. Indeed, the operations of every organization depend on the ability to share information. Despite this, few organizations manage these assets with rigor. Within most organizations, data disparity (differences in data structure, format, and use of values) is a larger problem than just simple errors; it can be a major obstacle to the integration of data. One of the reasons data stewardship programs focus on defining terms and consolidating the language around data is because that is the starting point for getting to more consistent data.

Many governance and information asset programs are driven solely by compliance, rather than by the potential value to be derived from data as an asset. A lack of recognition on the part of leadership means a lack of commitment within an organization to managing data as an asset, including managing its quality (Evans and Price, 2012). (See Figure 94.)

Barriers to effective management of data quality include:⁸⁵

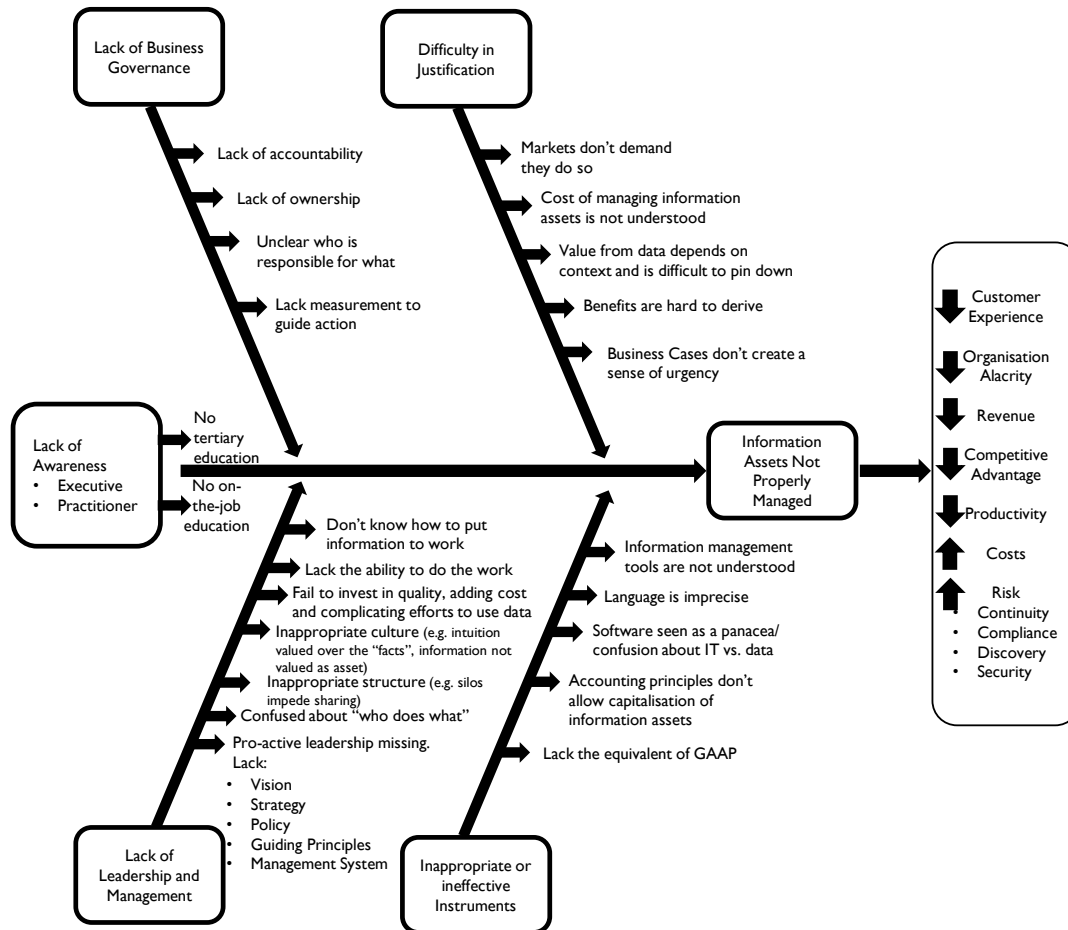
- Lack of awareness on the part of leadership and staff
- Lack of business governance
- Lack of leadership and management
- Difficulty in justification of improvements
- Inappropriate or ineffective instruments to measure value

These barriers have negative effects on customer experience, productivity, morale, organizational effectiveness, revenue, and competitive advantage. They increase costs of running the organization and introduce risks as well. (See Chapter 11.)

1.3.8.2 Issues Caused by Data Entry Processes

- **Data entry interface issues:** Poorly designed data entry interfaces can contribute to data quality issues. If a data entry interface does not have edits or controls to prevent incorrect data from being put in the system data processors are likely to take shortcuts, such as skipping non-mandatory fields and failing to update defaulted fields.
- **List entry placement:** Even simple features of data entry interfaces, such as the order of values within a drop-down list, can contribute to data entry errors.
- **Field overloading:** Some organizations re-use fields over time for different business purposes rather than making changes to the data model and user interface. This practice results in inconsistent and confusing population of the fields.
- **Training issues:** Lack of process knowledge can lead to incorrect data entry, even if controls and edits are in place. If data processors are not aware of the impact of incorrect data or if they are incented for speed, rather than accuracy, they are likely to make choices based on drivers other than the quality of the data.

⁸⁵ Adapted from *The Leader's Data Manifesto*. <https://dataleaders.org/>.



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Barriers that slow/hinder/prevent companies from managing their information as a business asset
Most commonly observed root causes
Danette McGilvray / James Price / Tom Redman
October 2016

Work based on research by Dr. Nina Evans and James Price, see
"Barriers to the Effective Deployment of Information Assets" at
www.dataleaders.org

Figure 94 Barriers to Managing Information as a Business Asset⁸⁶

- **Changes to business processes:** Business processes change over time, and with these changes new business rules and data quality requirements are introduced. However, business rule changes are not always incorporated into systems in a timely manner or comprehensively. Data errors will result if an interface is not upgraded to accommodate new or changed requirements. In addition, data is likely to be impacted unless changes to business rules are propagated throughout the entire system.

⁸⁶ Diagram developed by Danette McGilvray, James Price, and Tom Redman. Used by permission. <https://dataleaders.org/>.

- **Inconsistent business process execution:** Data created through processes that are executed inconsistently is likely to be inconsistent. Inconsistent execution may be due to training or documentation issues as well as to changing requirements.

1.3.8.3 Issues Caused by Data Processing Functions

- **Incorrect assumptions about data sources:** Production issues can occur due to errors or changes, inadequate or obsolete system documentation, or inadequate knowledge transfer (for example, when SMEs leave without documenting their knowledge). System consolidation activities, such as those associated with mergers and acquisitions, are often based on limited knowledge about the relationship between systems. When multiple source systems and data feeds need to be integrated there is always a risk that details will be missed, especially with varying levels of source knowledge available and tight timelines.
- **Stale business rules:** Over time, business rules change. They should be periodically reviewed and updated. If there is automated measurement of rules, the technical process for measuring rules should also be updated. If it is not updated, issues may not be identified or false positives will be produced (or both).
- **Changed data structures:** Source systems may change structures without informing downstream consumers (both human and system) or without providing sufficient time to account for the changes. This can result in invalid values or other conditions that prevent data movement and loading, or in more subtle changes that may not be detected immediately.

1.3.8.4 Issues Caused by System Design

- **Failure to enforce referential integrity:** Referential integrity is necessary to ensure high quality data at an application or system level. If referential integrity is not enforced or if validation is switched off (for example, to improve response times), various data quality issues can arise:
 - Duplicate data that breaks uniqueness rules
 - Orphan rows, which can be included in some reports and excluded from others, leading to multiple values for the same calculation
 - Inability to upgrade due to restored or changed referential integrity requirements
 - Inaccurate data due to missing data being assigned default values
- **Failure to enforce uniqueness constraints:** Multiple copies of data instances within a table or file expected to contain unique instances. If there are insufficient checks for uniqueness of instances, or if the unique constraints are turned off in the database to improve performance, data aggregation results can be overstated.

- **Coding inaccuracies and gaps:** If the data mapping or layout is incorrect, or the rules for processing the data are not accurate, the data processed will have data quality issues, ranging from incorrect calculations to data being assigned to or linked to improper fields, keys, or relationships.
- **Data model inaccuracies:** If assumptions within the data model are not supported by the actual data, there will be data quality issues ranging from data loss due to field lengths being exceeded by the actual data, to data being assigned to improper IDs or keys.
- **Field overloading:** Re-use of fields over time for different purposes, rather than changing the data model or code can result in confusing sets of values, unclear meaning, and potentially, structural problems, like incorrectly assigned keys.
- **Temporal data mismatches:** In the absence of a consolidated data dictionary, multiple systems could implement disparate date formats or timings, which in turn lead to data mismatch and data loss when data synchronization takes place between different source systems.
- **Weak Master Data Management:** Immature Master Data Management can lead to choosing unreliable sources for data, which can cause data quality issues that are very difficult to find until the assumption that the data source is accurate is disproved.
- **Data duplication:** Unnecessary data duplication is often a result of poor data management. There are two main types of undesirable duplication issues:
 - **Single Source – Multiple Local Instances:** For example, instances of the same customer in multiple (similar or identical) tables in the same database. Knowing which instance is the most accurate for use can be difficult without system-specific knowledge.
 - **Multiple Sources – Single Instance:** Data instances with multiple authoritative sources or systems of record. For example, single customer instances coming from multiple point-of-sale systems. When processing this data for use, there can be duplicate temporary storage areas. Merge rules determine which source has priority over others when processing into permanent production data areas.

1.3.8.5 Issues Caused by Fixing Issues

Manual data patches are changes made directly on the data in the database, not through the business rules in the application interfaces or processing. These are scripts or manual commands generally created in a hurry and used to ‘fix’ data in an emergency such as intentional injection of bad data, lapse in security, internal fraud, or external source for business disruption.

Like any untested code, they have a high risk of causing further errors through unintended consequences, by changing more data than required, or not propagating the patch to all historical data affected by the original issue. Most such patches also change the data in place, rather than preserving the prior state and adding corrected rows.

These changes are generally NOT undo-able without a complete restore from backup as there is only the database log to show the changes. Therefore, these shortcuts are strongly discouraged – they are opportunities for security breaches and business disruption longer than a proper correction would cause. All changes should go through a governed change management process.

1.3.9 Data Profiling

Data Profiling is a form of data analysis used to inspect data and assess quality. Data profiling uses statistical techniques to discover the true structure, content, and quality of a collection of data (Olson, 2003). A profiling engine produces statistics that analysts can use to identify patterns in data content and structure. For example:

- **Counts of nulls:** Identifies nulls exist and allows for inspection of whether they are allowable or not
- **Max/Min value:** Identifies outliers, like negatives
- **Max/Min length:** Identifies outliers or invalids for fields with specific length requirements
- **Frequency distribution** of values for individual columns: Enables assessment of reasonability (e.g., distribution of country codes for transactions, inspection of frequently or infrequently occurring values, as well as the percentage of the records populated with defaulted values)
- **Data type and format:** Identifies level of non-conformance to format requirements, as well as identification of unexpected formats (e.g., number of decimals, embedded spaces, sample values)

Profiling also includes cross-column analysis, which can identify overlapping or duplicate columns and expose embedded value dependencies. Inter-table analysis explores overlapping values sets and helps identify foreign key relationships. Most data profiling tools allow for drilling down into the analyzed data for further investigation.

Results from the profiling engine must be assessed by an analyst to determine whether data conforms to rules and other requirements. A good analyst can use profiling results to confirm known relationships and uncover hidden characteristics and patterns within and between data sets, including business rules, and validity constraints. Profiling is usually used as part of data discovery for projects (especially data integration projects; see Chapter 8) or to assess the current state of data that is targeted for improvement. Results of data profiling can be used to identify opportunities to improve the quality of both data and Metadata (Olson, 2003; Maydanchik, 2007).

While profiling is an effective way to understand data, it is just a first step to data quality improvement. It enables organizations to identify potential problems. Solving problems requires other forms of analysis, including business process analysis, analysis of data lineage, and deeper data analysis that can help isolate root causes of problems.

1.3.10 Data Quality and Data Processing

While the focus of data quality improvement efforts is often on the prevention of errors, data quality can also be improved through some forms of data processing. (See Chapter 8.)

1.3.10.1 Data Cleansing

Data Cleansing or *Scrubbing* transforms data to make it conform to data standards and domain rules. Cleansing includes detecting and correcting data errors to bring the quality of data to an acceptable level.

It costs money and introduces risk to continuously remediate data through cleansing. Ideally, the need for data cleansing should decrease over time, as root causes of data issues are resolved. The need for data cleansing can be addressed by:

- Implementing controls to prevent data entry errors
- Correcting the data in the source system
- Improving the business processes that create the data

In some situations, correcting on an ongoing basis may be necessary, as re-processing the data in a midstream system is cheaper than any other alternative.

1.3.10.2 Data Enhancement

Data enhancement or enrichment is the process of adding attributes to a data set to increase its quality and usability. Some enhancements are gained by integrating data sets internal to an organization. External data can also be purchased to enhance organizational data (see Chapter 10). Examples of data enhancement include:

- **Time/Date stamps:** One way to improve data is to document the time and date that data items are created, modified, or retired, which can help to track historical data events. If issues are detected with the data, timestamps can be very valuable in root cause analysis, because they enable analysts to isolate the timeframe of the issue.
- **Audit data:** Auditing can document data lineage, which is important for historical tracking as well as validation.
- **Reference vocabularies:** Business specific terminology, ontologies, and glossaries enhance understanding and control while bringing customized business context.
- **Contextual information:** Adding context such as location, environment, or access methods and tagging data for review and analysis.
- **Geographic information:** Geographic information can be enhanced through address standardization and geocoding, which includes regional coding, municipality, neighborhood mapping, latitude / longitude pairs, or other kinds of location-based data.
- **Demographic information:** Customer data can be enhanced through demographic information, such as age, marital status, gender, income, or ethnic coding. Business entity data can be associated with annual revenue, number of employees, size of occupied space, etc.

- **Psychographic information:** Data used to segment the target populations by specific behaviors, habits, or preferences, such as product and brand preferences, organization memberships, leisure activities, commuting transportation style, shopping time preferences, etc.
- **Valuation information:** Use this kind of enhancement for asset valuation, inventory, and sale.

1.3.10.3 Data Parsing and Formatting

Data Parsing is the process of analyzing data using pre-determined rules to define its content or value. Data parsing enables the data analyst to define sets of patterns that feed into a rule engine used to distinguish between valid and invalid data values. Matching specific pattern(s) triggers actions.

Data parsing assigns characteristics to the data values appearing in a data instance, and those characteristics help in determining potential sources for added benefits. For example, if an attribute called ‘name’ can be determined to have values belonging to ‘business name’ embedded within it, then the data value is identified as the name of a business rather than the name of a person. Use the same approach for any situation in which data values organize into semantic hierarchies such as sub-parts, parts, and assemblies.

Many data quality issues involve situations where variation in data values representing similar concepts introduces ambiguity. Extract and rearrange the separate components (commonly referred to as ‘tokens’) can be extracted and rearranged into a standard representation to create a valid pattern. When an invalid pattern is recognized, the application may attempt to transform the invalid value into one that meets the rules. Perform standardization by mapping data from some source pattern into a corresponding target representation.

For example, consider the different ways telephone numbers expected to conform to a numbering plan are formatted. While some have digits, some have alphabetic characters, and all use different special characters for separation. People can recognize each one as a telephone number. However, to determine if these numbers are accurate (perhaps by comparing them to a master customer directory), or to investigate whether duplicate numbers exist when there should be only one for each supplier, the values must be parsed into their component segments (area code, exchange, and line number) and then transformed into a standard format.

Another good example is a customer name, since names may be represented in thousands of different forms. A good standardization tool will be able to parse the different components of a customer name, such as given name, middle name, family name, initials, titles, generational designations, and then rearrange those components into a canonical representation that other data services will be able to manipulate.

The human ability to recognize familiar patterns contributes to an ability to characterize variant data values belonging to the same abstract class of values; people recognize different types of telephone numbers because they conform to frequently used patterns. An analyst describes the format patterns that all represent a data object, such as **Person Name**, **Product Description**, and so on. A data quality tool parses data values that conform to any of those patterns, and even transforms them into a single, standardized form that will simplify the assessment, similarity analysis, and remediation processes. Pattern-based parsing can automate the recognition and subsequent standardization of meaningful value components.

1.3.10.4 Data Transformation and Standardization

During normal processing, data rules trigger and transform the data into a format that is readable by the target architecture. However, readable does not always mean acceptable. Rules are created directly within a data integration stream, or rely on alternate technologies embedded in or accessible from within a tool.

Data transformation builds on these types of standardization techniques. Guide rule-based transformations by mapping data values in their original formats and patterns into a target representation. Parsed components of a pattern are subjected to rearrangement, corrections, or any changes as directed by the rules in the knowledge base. In fact, standardization is a special case of transformation, employing rules that capture context, linguistics, and idioms recognized as common over time, through repeated analysis by the rules analyst or tool vendor. (See Chapter 3.)

2. Activities

2.1 Define High Quality Data

Many people recognize poor quality data when they see it. Fewer are able to define what they mean by high quality data. Alternatively, they define it in very general term: “The data has to be right.” “We need accurate data.” High quality data is fit for the purposes of data consumers. Before launching a Data Quality program, it is beneficial to understand business needs, define terms, identify organizational pain points, and start to build consensus about the drivers and priorities for data quality improvement. Ask a set of questions to understand current state and assess organizational readiness for data quality improvement:

- What do stakeholders mean by ‘high quality data’?
- What is the impact of low quality data on business operations and strategy?
- How will higher quality data enable business strategy?
- What priorities drive the need for data quality improvement?
- What is the tolerance for poor quality data?
- What governance is in place to support data quality improvement?
- What additional governance structures will be needed?

Getting a comprehensive picture of the current state of data quality in an organization requires approaching the question from different perspectives:

- An understanding of business strategy and goals
- Interviews with stakeholders to identify pain points, risks, and business drivers
- Direct assessment of data, through profiling and other form of analysis
- Documentation of data dependencies in business processes
- Documentation of technical architecture and systems support for business processes

This kind of assessment can reveal a significant number of opportunities. These need to be prioritized based on the potential benefit to the organization. Using input from stakeholders, including Data Stewards and business and technical SMEs, the Data Quality team should define the meaning of data quality and propose program priorities.

2.2 Define a Data Quality Strategy

Improving data quality requires a strategy that accounts for the work that needs to be done and the way people will execute it. Data quality priorities must align with business strategy. Adopting or developing a framework and methodology will help guide both strategy and tactics while providing a means to measure progress and impacts. A framework should include methods to:

- Understand and prioritize business needs
- Identify the data critical to meeting business needs
- Define business rules and data quality standards based on business requirements
- Assess data against expectations
- Share findings and get feedback from stakeholders
- Prioritize and manage issues
- Identify and prioritize opportunities for improvement
- Measure, monitor, and report on data quality
- Manage Metadata produced through data quality processes
- Integrate data quality controls into business and technical processes

A framework should also account for how to organize for data quality and how to leverage data quality tools. As noted in the chapter introduction, improving data quality requires a Data Quality program team to engage business and technical staff and define a program of work that addresses critical issues, defines best practices, and puts in place operational processes that support ongoing management of data quality. Often such a team will be part of the Data Management Organization. DQ analysts will need to work closely with Data Stewards at all levels. They should also influence policy, including policy about business processes and systems development. However, such a team will not be able to solve all of an organization's data quality challenges. DQ work and a commitment to high quality data need to become embedded in organizational practices. The DQ Strategy should account for how to extend best practices. (See Chapter 17.)

2.3 Identify Critical Data and Business Rules

Not all data is of equal importance. Data Quality Management efforts should focus first on the most important data in the organization: data that, if it were of higher quality, would provide greater value to the organization and its customers. Data can be prioritized based on factors such as regulatory requirements, financial value, and direct impact on customers. Often, data quality improvement efforts start with Master Data, which is, by definition, among the most important data in any organization. The result of the importance analysis is a ranked list of data, which the Data Quality team can use to focus their work efforts.

Having identified the critical data, Data Quality analysts need to identify business rules that describe or imply expectations about the quality characteristics of data. Often rules themselves are not explicitly documented. They may need to be reverse-engineered through analysis of existing business processes, workflows, regulations, policies, standards, system edits, software code, triggers and procedures, status code assignment and use, and plain old common sense. For example, if a marketing company wants to target efforts at people in a specific demographic, then potential indexes of data quality might be the level and reasonability of population in demographic fields like birth date, age, gender, and household income.

Most business rules are associated with how data is collected or created, but data quality measurement centers around whether data is fit for use. The two (data creation and data use) are related. People want to use data because of what it represents and why it was created. For example, to understand an organization's sales performance during a specific quarter or over time depends on having reliable data about the sales process (number and type of units sold, volume sold to existing customers vs. new customers, etc.).

It is not possible to know all the ways that data might be used, but it is possible to understand the process and rules by which data was created or collected. Measurements that describe whether data is fit for use should be developed in relation to known uses and measurable rules based on dimensions of data quality: completeness, conformity, validity, integrity, etc. that provide the basis for meaningful metrics. Dimensions of quality enable analysts to characterize both rules (field X is mandatory and must be populated) and findings (e.g., the field is not populated in 3% of the records; the data is only 97% complete).

At the field or column level, rules can be straightforward. Completeness rules are a reflection of whether a field is mandatory or optional, and, if optional, the conditions under which it should be populated. Validity rules are dependent on stipulating the domain of valid values and, in some cases, the relationship between fields. For example, a US ZIP Code needs to be valid, in and of itself, and correctly associated with a US State code. Rules should also be defined at the data set level. For example, every customer must have a valid mailing address.

Defining data quality rules is challenging because most people are not used to thinking about data in terms of rules. It may be necessary to get at the rules indirectly, by asking stakeholders about the input and output requirements of a business process. It also helps to ask about pain points, what happens when data is missing or incorrect, how they identify issues, how they recognize bad data, etc. Keep in mind that it is not necessary to know all the rules in order to assess data. Discovery and refinement of rules is an ongoing process. One of the best ways to get at rules is to share results of assessments. These results often give stakeholders a new perspective on the data from which they can articulate rules that tell them what they need to know about the data.

2.4 Perform an Initial Data Quality Assessment

Once the most critical business needs and the data that supports them have been identified, the most important part of the data quality assessment is actually looking at that data, querying it to understand data content and relationships, and comparing actual data to rules and expectations. The first time this is done, analysts will discover many things: undocumented relationships and dependencies within the data, implied rules, redundant

data, contradictory data, etc., as well as data that actually does conform to rules. With the help of data stewards, other SMEs, and data consumers, DQ analysts will need to sort out and prioritize findings.

The goal of an initial data quality assessment is to learn about the data in order to define an actionable plan for improvement. It is usually best to start with a small, focused effort – a basic proof of concept – to demonstrate how the improvement process works. Steps include:

- Define the goals of the assessment; these will drive the work
- Identify the data to be assessed; focus should be on a small data set, even a single data element, or a specific data quality problem
- Identify uses of the data and the consumers of the data
- Identify known risks with the data to be assessed, including the potential impact of data issues on organizational processes
- Inspect the data based on known and proposed rules
- Document levels of non-conformance and types of issues
- Perform additional, in-depth analysis based on initial findings in order to
 - Quantify findings
 - Prioritize issues based on business impact
 - Develop hypotheses about root causes of data issues
- Meet with Data Stewards, SMEs, and data consumers to confirm issues and priorities
- Use findings as a foundation for planning
 - Remediation of issues, ideally at their root causes
 - Controls and process improvements to prevent issues from recurring
 - Ongoing controls and reporting

2.5 Identify and Prioritize Potential Improvements

Having proven that the improvement process can work, the next goal is to apply it strategically. Doing so requires identifying and prioritizing potential improvements. Identification may be accomplished by full-scale data profiling of larger data sets to understand the breadth of existing issues. It may also be accomplished by other means, such as interviewing stakeholders about the data issues that impact them and following up with analysis of the business impact of those issues. Ultimately, prioritization requires a combination of data analysis and discussion with stakeholders.

The steps to perform a full data profiling and analysis are essentially the same as those in performing a small-scale assessment: define goals, understand data uses and risks, measure against rules, document and confirm findings with SMEs, use this information to prioritize remediation and improvement efforts. However, there are sometimes technical obstacles to full-scale profiling. And the effort will need to be coordinated across a team of analysts and overall results will need to be summarized and understood if an effective action plan is to be put in place. Large-scale profiling efforts, like those on a smaller scale, should still focus on the most critical data.

Profiling data is only the first step in analysis of data quality issues. It helps identify issues, but does not identify root causes, nor does it determine the impact of issues to business processes. Determining impact requires input

from stakeholders along the data chain. When planning large scale profiling, ensure that time is allocated to share results, prioritize problems, and determine which issues require in-depth analysis.

2.6 Define Goals for Data Quality Improvement

The knowledge obtained through the preliminary assessments forms the basis for specific Data Quality program goals. Improvement can take different forms, from simple remediation (e.g., correction of errors on records) to remediation of root causes. Remediation and improvement plans should account for quick hits – issues that can be addressed immediately at low cost – and longer-term strategic changes. The strategic focus of such plans should be to address root causes of issues and to put in place mechanisms to prevent issues in the first place.

Be aware that many things can get in the way of improvement efforts: system constraints, age of data, ongoing project work that uses the questionable data, overall complexity of the data landscape, cultural resistance to change. To prevent these constraints from stalling the program, set specific, achievable goals based on consistent quantification of the business value of the improvements to data quality.

For example, a goal may be to improve the completeness of customer data from 90% to 95% based on process improvements and system edits. Obviously, showing improvement will involve comparing initial measurements and improved results. But the value comes with benefits of the improvement: fewer customer complaints, less time spent correcting errors, etc. Measure these things to explain the value of the improvement work. No one cares about levels of field completeness unless there is a business impact. There must be a positive return on investment for improvements to data. When issues are found, determine ROI of fixes based on:

- The criticality (importance ranking) of the data affected
- Amount of data affected
- The age of the data
- Number and type of business processes impacted by the issue
- Number of customers, clients, vendors, or employees impacted by the issue
- Risks associated with the issue
- Costs of remediating root causes
- Costs of potential work-arounds

In assessing issues, especially those where root causes are identified and technical changes are required, always seek out opportunities to prevent issues from recurring. Preventing issues generally costs less than correcting them – sometimes orders of magnitude less. (See Chapter 11.)

2.7 Develop and Deploy Data Quality Operations

Many Data Quality programs get started through a set of improvement projects identified via results of the data quality assessment. In order to sustain data quality, a DQ program should put in place a plan that allows the team to manage data quality rules and standards, monitor data's ongoing conformance with rules, identify and manage data quality issues, and report on quality levels. In support of these activities, DQ analysts and Data

Stewards will also be engaged in activities such as documenting data standards and business rules and establishing data quality requirements for vendors.

2.7.1 Manage Data Quality Rules

The process of profiling and analyzing data will help an organization discover (or reverse engineer) business and data quality rules. As the data quality practice matures, the capture of such rules should be built into the system development and enhancement process. Defining rules upfront will:

- Set clear expectations for data quality characteristics
- Provide requirements for system edits and controls that prevent data issues from being introduced
- Provide data quality requirements to vendors and other external parties
- Create the foundation for ongoing data quality measurement and reporting

In short, data quality rules and standards are a critical form of Metadata. To be effective, they need to be managed as Metadata. Rules should be:

- **Documented consistently:** Establish standards and templates for documenting rules so that they have a consistent format and meaning.
- **Defined in terms of Data Quality dimensions:** Dimensions of quality help people understand what is being measured. Consistent application of dimensions will help with the measurement and issue management processes.
- **Tied to business impact:** While data quality dimensions enable understanding of common problems, they are not a goal in-and-of-themselves. Standards and rules should be connected directly to their impact on organizational success. Measurements that are not tied to business processes should not be taken.
- **Backed by data analysis:** Data Quality Analysts should not guess at rules. Rules should be tested against actual data. In many cases, rules will show that there are issues with the data. But analysis can also show that the rules themselves are not complete.
- **Confirmed by SMEs:** The goal of the rules is to describe how the data should look. Often, it takes knowledge of organizational processes to confirm that rules correctly describe the data. This knowledge comes when subject matter experts confirm or explain the results of data analysis.
- **Accessible to all data consumers:** All data consumers should have access to documented rules. Such access allows them to better understand the data. It also helps to ensure that the rules are correct and complete. Ensure that consumers have a means to ask questions about and provide feedback on rules.

2.7.2 Measure and Monitor Data Quality

The operational Data Quality Management procedures depend on the ability to measure and monitor the quality of data. There are two equally important reasons to implement operational data quality measurements:

- To inform data consumers about levels of quality
- To manage risk that change may be introduced through changes to business or technical processes

Some measurements serve both purposes. Measurements should be developed based on findings from data assessment and root cause analysis. Measurements intended to inform data consumers will focus on critical data elements and relationships that, if they are not sound, will directly impact business processes. Measurements related to managing risk should focus on relationships that have gone wrong in the past and may go wrong in the future. For example, if data is derived based on a set of ETL rules and those rules may be impacted by changes to business processes, measurements should be put in place to detect changes to the data.

Knowledge of past problems should be applied to manage risk. For example, if numerous data issues are associated with complex derivations, then all derivations should be assessed – even those that have not been associated with data issues. In most cases, it is worthwhile to put in place measurements that monitor functions similar to those that have had problems.

Measurement results can be described at two levels: the detail related to the execution of individual rules and overall results aggregated from the rules. Each rule should have a standard, target, or threshold index for comparison. This function most often reflects the percentage of correct data or percentage of exceptions depending on the formula used. For example:

$$\text{ValidDQI}(r) = \frac{(\text{TestExecutions}(r) - \text{ExceptionsFound}(r))}{\text{TestExecutions}(r)}$$

$$\text{InvalidDQI}(r) = \frac{(\text{ExceptionsFound}(r))}{\text{TestExecutions}(r)}$$

R represents the rule being tested. For example, 10,000 tests of a business rule (r) found 560 exceptions. In this example, the ValidDQ result would be $9440/10,000 = 94.4\%$, and the Invalid DQ result would be $560/10,000 = 5.6\%$.

Organizing the metrics and results as shown in Table 30 can help to structure measures, metrics, and indicators across the report, reveal possible rollups, and enhance communications. The report can be more formalized and linked to projects that will remediate the issues. Filtered reports are useful for data stewards looking for trends and contributions. Table 30 provides examples of rules constructed in this manner. Where applicable, results of rules are expressed in both positive percentages (the portion of the data that conforms to rules and expectations) and negative percentages (the portion of the data that does not conform to the rule).

Data quality rules provide the foundation for operational management of data quality. Rules can be integrated into application services or data services that supplement the data lifecycle, either through Commercial Off The Shelf (COTS) data quality tools, rules engines and reporting tools for monitoring and reporting, or custom-developed applications.

Table 30 DQ Metric Examples

Dimension and Business Rule	Measure	Metrics	Status Indicator
Completeness Business Rule 1: Population of field is mandatory	Count the number of records where data is populated, compare to the total number of records	Divide the obtained number of records where data is populated by the total number of records in the table or database and multiply it by 100 to get to percentage complete	Unacceptable: Below 80% populated Above 20% not populated
Example 1: Postal Code must be populated in the address table	Count populated: 700,000 Count not populated: 300,000 Total count: 1,000,000	Positive measure: $700,000/1,000,000 \times 100 = 70\%$ populated Negative measure: $300,000/1,000,000 \times 100 = 30\%$ not populated	Example result: Unacceptable
Uniqueness Business Rule 2: There should be only one record per entity instance in a table	Count the number of duplicate records identified; report on the percentage of records that represent duplicates	Divide the number of duplicate records by the total number of records in the table or database and multiply it by 100	Unacceptable: Above 0%
Example 2: There should be one and only one current row per postal code on the Postal Codes master list	Count of duplicates: 1,000 Total Count: 1,000,000	$10,000/1,000,000 \times 100 = 1.0\%$ of postal codes are present on more than one current row	Example result: Unacceptable
Timeliness Business Rule 3: Records must arrive within a scheduled timeframe	Count the number of records failing to arrive on time from a data service for business transactions to be completed	Divide the number of incomplete transactions by the total number of attempted transactions in a time period and multiply by 100	Unacceptable: Below 99% completed on time Above 1% not completed on time
Example 3: Equity market record should arrive within 5 minutes of being transacted	Count of incomplete transactions: 2000 Count of attempted transactions: 1,000,000	Positive: $(1,000,000 - 2000) / 1,000,000 \times 100 = 99.8\%$ of transaction records arrived within defined timeframe Negative: $2000/1,000,000 \times 100 = 0.20\%$ of transactions did not arrive within defined timeframe	Example Result: Acceptable
Validity Business Rule 4: If field X = value 1, then field Y must = value 1-prime	Count the number of records where the rule is met	Divide the number of records that meet the condition by the total number of records	Unacceptable : Below 100% adherence to the rule
Example 4: Only shipped orders should be billed	Count of records where status for shipping = Shipped and status for billing = Billed: 999,000 Count of total records: 1,000,000	Positive: $999,000/1,000,000 \times 100 = 99.9\%$ of records conform to the rule Negative: $(1,000,000 - 999,000) / 1,000,000 \times 100 = 0.10\%$ do not conform to the rule	Example Result: Unacceptable

Provide continuous monitoring by incorporating control and measurement processes into the information processing flow. Automated monitoring of conformance to data quality rules can be done in-stream or through a batch process. Measurements can be taken at three levels of granularity: the data element value, data instance or record, or the data set. Table 31 describes techniques for collecting data quality measurements. In-stream measurements can be taken while creating data or handing data off between processing stages. Batch queries can be performed on collections of data instances assembled in a data set, usually in persistent storage. Data set measurements generally cannot be taken in-stream, since the measurement may need the entire set.

Incorporating the results of the control and measurement processes into both the operational procedures and reporting frameworks enables continuous monitoring of the levels of data quality for feedback and improvement to the data generation/collection activities.

Table 31 Data Quality Monitoring Techniques

Granularity	In-stream (In-Process Flow) Treatment	Batch Treatment
Data Element	Edit checks in application Data element validation services Specially programmed applications	Direct queries Data profiling or analyzer tool
Data Record	Edit checks in application Data record validation services Specially programmed applications	Direct queries Data profiling or analyzer tool
Data set	Inspection inserted between processing stages	Direct queries Data profiling or analyzer tool

2.7.3 Develop Operational Procedures for Managing Data Issues

Whatever tools are used to monitor data quality, when results are assessed by Data Quality team members, they need to respond to findings in a timely and effective manner. The team must design and implement detailed operational procedures for:

- **Diagnosing issues:** The objective is to review the symptoms of the data quality incident, trace the lineage of the data in question, identify the problem and where it originated, and pinpoint potential root causes of the problem. The procedure should describe how the Data Quality Operations team would:
 - Review the data issues in the context of the appropriate information processing flows and isolate the location in the process where the flaw is introduced
 - Evaluate whether there have been any environmental changes that would cause errors entering into the system
 - Evaluate whether or not there are any other process issues that contributed to the data quality incident
 - Determine whether there are issues with external data that have affected the quality of the data

NOTE: The work of root cause analysis requires input from technical and business SMEs. While the DQ team may lead and facilitate this kind of work effort, success requires cross-functional collaboration

- **Formulating options for remediation:** Based on the diagnosis, evaluate alternatives for addressing the issue. These may include:
 - Addressing non-technical root causes such as lack of training, lack of leadership support, unclear accountability and ownership, etc.
 - Modification of the systems to eliminate technical root causes
 - Developing controls to prevent the issue
 - Introducing additional inspection and monitoring
 - Directly correcting flawed data
 - Taking no action based on the cost and impact of correction versus the value of the data correction
- **Resolving issues:** Having identified options for resolving the issue, the Data Quality team must confer with the business data owners to determine the best way to resolve the issue. These procedures should detail how the analysts:
 - Assess the relative costs and merits of the alternatives
 - Recommend one of the planned alternatives
 - Provide a plan for developing and implementing the resolution
 - Implement the resolution

Decisions made during the issue management process should be tracked in an incident tracking system. When the data in such a system is managed well, it can provide valuable insight about the causes and costs of data issues. Include a description of the issue and the root causes, options for remediation, and the decision on how to resolve the issue.

The incident tracking system will collect performance data relating to issue resolution, work assignments, volume of issues, frequency of occurrence, as well as the time to respond, diagnose, plan a solution, and resolve issues. These metrics can provide valuable insights into the effectiveness of the current workflow, as well as systems and resource utilization, and they are important management data points that can drive continuous operational improvement for data quality control.

Incident tracking data also helps data consumers. Decisions based upon remediated data should be made with knowledge that it has been changed, why it has been changed, and how it has been changed. That is one reason why it is important to record the methods of modification and the rationale for them. Make this documentation available to data consumers and developers researching code changes. While changes may be obvious to the people who implement them, the history of changes will be lost to future data consumers unless it is documented. Data quality incident tracking requires staff be trained on how issues should be classified, logged, and tracked. To support effective tracking:

- **Standardize data quality issues and activities:** Since the terms used to describe data issues may vary across lines of business, it is valuable to define a standard vocabulary for the concepts used. Doing so will simplify classification and reporting. Standardization also makes it easier to measure the volume of issues and activities, identify patterns and interdependencies between systems and participants, and

report on the overall impact of data quality activities. The classification of an issue may change as the investigation deepens and root causes are exposed.

- **Provide an assignment process for data issues:** The operational procedures direct the analysts to assign data quality incidents to individuals for diagnosis and to provide alternatives for resolution. Drive the assignment process within the incident tracking system by suggesting those individuals with specific areas of expertise.
- **Manage issue escalation procedures:** Data quality issue handling requires a well-defined system of escalation based on the impact, duration, or urgency of an issue. Specify the sequence of escalation within the data quality Service Level Agreement. The incident tracking system will implement the escalation procedures, which helps expedite efficient handling and resolution of data issues.
- **Manage data quality resolution workflow:** The data quality SLA specifies objectives for monitoring, control, and resolution, all of which define a collection of operational workflows. The incident tracking system can support workflow management to track progress with issues diagnosis and resolution.

2.7.4 Establish Data Quality Service Level Agreements

A data quality Service Level Agreement (SLA) specifies an organization's expectations for response and remediation for data quality issues in each system. Data quality inspections as scheduled in the SLA help to identify issues to fix, and over time, reduce the number of issues. While enabling the isolation and root cause analysis of data flaws, there is an expectation that the operational procedures will provide a scheme for remediation of root causes within an agreed timeframe. Having data quality inspection and monitoring in place increases the likelihood of detection and remediation of a data quality issue before a significant business impact can occur. Operational data quality control defined in a data quality SLA includes:

- Data elements covered by the agreement
- Business impacts associated with data flaws
- Data quality dimensions associated with each data element
- Expectations for quality for each data element for each of the identified dimensions in each application or system in the data value chain
- Methods for measuring against those expectations
- Acceptability threshold for each measurement
- Steward(s) to be notified in case the acceptability threshold is not met
- Timelines and deadlines for expected resolution or remediation of the issue
- Escalation strategy, and possible rewards and penalties

The data quality SLA also defines the roles and responsibilities associated with performance of operational data quality procedures. The operational data quality procedures provide reports in conformance with the defined business rules, as well as monitoring staff performance in reacting to data quality incidents. Data stewards and the operational data quality staff, while upholding the level of data quality service, should consider their data quality SLA constraints and connect data quality to individual performance plans.

When issues are not addressed within the specified resolution times, an escalation process must exist to communicate non-observance of the level of service up the management and governance chain. The data quality SLA establishes the time limits for notification generation, the names of those in that management chain, and when escalation needs to occur. Given the set of data quality rules, methods for measuring conformance, the acceptability thresholds defined by the business clients, and the service level agreements, the Data Quality team can monitor compliance of the data to the business expectations, as well as how well the Data Quality team performs on the procedures associated with data errors.

SLA reporting can be on a scheduled basis driven by business and operational requirements. Particular focus will be on report trend analysis in cases focused on periodic rewards and penalties if such concepts are built into the SLA framework.

2.7.5 Develop Data Quality Reporting

The work of assessing the quality of data and managing data issues will not benefit the organization unless the information is shared through reporting so that data consumers understand the condition of the data. Reporting should focus around:

- Data quality scorecard, which provides a high-level view of the scores associated with various metrics, reported to different levels of the organization within established thresholds
- Data quality trends, which show over time how the quality of data is measured, and whether trending is up or down
- SLA Metrics, such as whether operational data quality staff diagnose and respond to data quality incidents in a timely manner
- Data quality issue management, which monitors the status of issues and resolutions
- Conformance of the Data Quality team to governance policies
- Conformance of IT and business teams to Data Quality policies
- Positive effects of improvement projects

Reporting should align to metrics in the data quality SLA as much as possible, so that the team's goals are aligned with those of its customers. The Data Quality program should also report on the positive effects of improvement projects. It is best to do this in business terms to continually remind the organization of the direct effect that data has on customers.

3. Tools

Tools should be selected and tool architectures should be set in the planning phase of the enterprise Data Quality program. Tools provide a partial rule set starter kit but organizations need to create and input their own context specific rules and actions into any tool.

3.1 Data Profiling Tools

Data profiling tools produce high-level statistics that enable analysts to identify patterns in data and perform initial assessment of quality characteristics. Some tools can be used to perform ongoing monitoring of data. Profiling tools are particularly important for data discovery efforts because they enable assessment of large data sets. Profiling tools augmented with data visualization capabilities will aid in the process of discovery. (See Chapters 5 and 8, and Section 1.3.9.)

3.2 Data Querying Tools

Data profiling is only the first step in data analysis. It helps identify potential issues. Data Quality team members also need to query data more deeply to answer questions raised by profiling results and find patterns that provide insight into root causes of data issues. For example, querying to discover and quantify other aspects of data quality, such as uniqueness and integrity.

3.3 Modeling and ETL Tools

The tools used to model data and create ETL processes have a direct impact on the quality of data. If used with the data in mind, these tools can enable higher quality data. If they are used without knowledge of the data, they can have detrimental effects. DQ team members should work with development teams to ensure that data quality risks are addressed and that the organization takes full advantage of the ways in which effective modeling and data processing can enable higher quality data. (See Chapters 5, 8, and 11.)

3.4 Data Quality Rule Templates

Rule templates allow analyst to capture expectations for data. Templates also help bridge the communications gap between business and technical teams. Consistent formulation of rules makes it easier to translate business needs into code, whether that code is embedded in a rules engine, the data analyzer component of a data-profiling tool, or a data integration tool. A template can have several sections, one for each type of business rule to implement.

3.5 Metadata Repositories

As noted in Section 1.3.4, defining data quality requires Metadata and definitions of high quality data are a valuable kind of Metadata. DQ teams should work closely with teams that manage Metadata to ensure that data quality requirements, rules, measurement results, and documentation of issues are made available to data consumers.

4. Techniques

4.1 Preventive Actions

The best way to create high quality data is to prevent poor quality data from entering an organization. Preventive actions stop known errors from occurring. Inspecting data after it is in production will not improve its quality. Approaches include:

- **Establish data entry controls:** Create data entry rules that prevent invalid or inaccurate data from entering a system.
- **Train data producers:** Ensure staff in upstream systems understand the impact of their data on downstream users. Give incentives or base evaluations on data accuracy and completeness, rather than just speed.
- **Define and enforce rules:** Create a ‘data firewall,’ which has a table with all the business data quality rules used to check if the quality of data is good, before being used in an application such as a data warehouse. A data firewall can inspect the level of quality of data processed by an application, and if the level of quality is below acceptable levels, analysts can be informed about the problem.
- **Demand high quality data from data suppliers:** Examine an external data provider’s processes to check their structures, definitions, and data source(s) and data provenance. Doing so enables assessment of how well their data will integrate and helps prevent the use of non-authoritative data or data acquired without permission from the owner.
- **Implement Data Governance and Stewardship:** Ensure roles and responsibilities are defined that describe and enforce rules of engagement, decision rights, and accountabilities for effective management of data and information assets (McGilvray, 2008). Work with data stewards to revise the process of, and mechanisms for, generating, sending, and receiving data.
- **Institute formal change control:** Ensure all changes to stored data are defined and tested before being implemented. Prevent changes directly to data outside of normal processing by establishing gating processes.

4.2 Corrective Actions

Corrective actions are implemented after a problem has occurred and been detected. Data quality issues should be addressed systemically and at their root causes to minimize the costs and risks of corrective actions. ‘Solve the problem where it happens’ is the best practice in Data Quality Management. This generally means that corrective actions should include preventing recurrence of the causes of the quality problems.

Perform data correction in three general ways:

- **Automated correction:** Automated correction techniques include rule-based standardization, normalization, and correction. The modified values are obtained or generated and committed without manual intervention. An example is automated address correction, which submits delivery addresses to an address standardizer that conforms and corrects delivery addresses using rules, parsing, standardization, and reference tables. Automated correction requires an environment with well-defined standards, commonly accepted rules, and known error patterns. The amount of automated correction can be reduced over time if this environment is well-managed and corrected data is shared with upstream systems.
- **Manually-directed correction:** Use automated tools to remediate and correct data but require manual review before committing the corrections to persistent storage. Apply name and address remediation, identity resolution, and pattern-based corrections automatically, and use some scoring mechanism to propose a level of confidence in the correction. Corrections with scores above a particular level of confidence may be committed without review, but corrections with scores below the level of confidence are presented to the data steward for review and approval. Commit all approved corrections, and review those not approved to understand whether to adjust the applied underlying rules. Environments in which sensitive data sets require human oversight (e.g., MDM) are good examples of where manual-directed correction may be suited.
- **Manual correction:** Sometimes manual correction is the only option in the absence of tools or automation or if it is determined that the change is better handled through human oversight. Manual corrections are best done through an interface with controls and edits, which provide an audit trail for changes. The alternative of making corrections and committing the updated records directly in production environments is extremely risky. Avoid using this method.

4.3 Quality Check and Audit Code Modules

Create shareable, linkable, and re-usable code modules that execute repeated data quality checks and audit processes that developers can get from a library. If the module needs to change, then all the code linked to that module will get updated. Such modules simplify the maintenance process. Well-engineered code blocks can prevent many data quality problems. As importantly, they ensure processes are executed consistently. Where laws or policy mandate reporting of specific quality results, the lineage of results often needs to be described. Quality check modules can provide this. For data that has any questionable quality dimension and that is highly rated, qualify the information in the shared environments with quality notes, and confidence ratings.

4.4 Effective Data Quality Metrics

A critical component of managing data quality is developing metrics that inform data consumers about quality characteristics that are important to their uses of data. Many things can be measured, but not all of them are worth the time and effort. In developing metrics, DQ analysts should account for these characteristics:

- **Measurability:** A data quality metric must be measurable – it needs to be something that can be counted. For example, data relevancy is not measurable, unless clear criteria are set for what makes data relevant. Even data completeness needs to be objectively defined in order to be measured. Expected results should be quantifiable within a discrete range.
- **Business relevance:** While many things are measurable, not all translate into useful metrics. Measurements need to be relevant to data consumers. The value of the metric is limited if it cannot be related to some aspect of business operations or performance. Every data quality metric should correlate with the influence of the data on key business expectations.
- **Acceptability:** The data quality dimensions frame the business requirements for data quality. Quantifying along the identified dimension provides hard evidence of data quality levels. Determine whether data meets business expectations based on specified acceptability thresholds. If the score is equal to or exceeds the threshold, the quality of the data meets business expectations. If the score is below the threshold, it does not.
- **Accountability / Stewardship:** Metrics should be understood and approved by key stakeholders (e.g., business owners and Data Stewards). They are notified when the measurement for the metric shows that the quality does not meet expectations. The business data owner is accountable, while a data steward takes appropriate corrective action.
- **Controllability:** A metric should reflect a controllable aspect of the business. In other words, if the metric is out of range, it should trigger action to improve the data. If there is no way to respond, then the metric is probably not useful.
- **Trending:** Metrics enable an organization to measure data quality improvement over time. Tracking helps Data Quality team members monitor activities within the scope of a data quality SLA and data sharing agreement, and demonstrate the effectiveness of improvement activities. Once an information process is stable, statistical process control techniques can be applied to detect changes to the predictability of the measurement results and the business and technical processes on which it provides insight.

4.5 Statistical Process Control

Statistical Process Control (SPC) is a method to manage processes by analyzing measurements of variation in process inputs, outputs, or steps. The technique was developed in the manufacturing sector in the 1920s and has been applied in other industries, in improvement methodologies such as Six Sigma, and in Data Quality Management.⁸⁷ Simply defined, a process is a series of steps executed to turn inputs into outputs. SPC is based on the assumption that when a process with consistent inputs is executed consistently, it will produce consistent outputs. It uses measures of central tendency (how values cluster around a central value, such as a mean,

⁸⁷ See Redman (1996 and 2001), Loshin (2000), Sebastian-Coleman (2013), Jugulum (2014).

median, or mode) and of variability around a central value (e.g., range, variance, standard deviation), to establish tolerances for variation within a process.

The primary tool used for SPC is the control chart (Figure 95), which is a time series graph that includes a central line for the average (the measure of central tendency) and depicts calculated upper and lower control limits (variability around a central value). In a stable process, measurement results outside the control limits indicate a special cause.

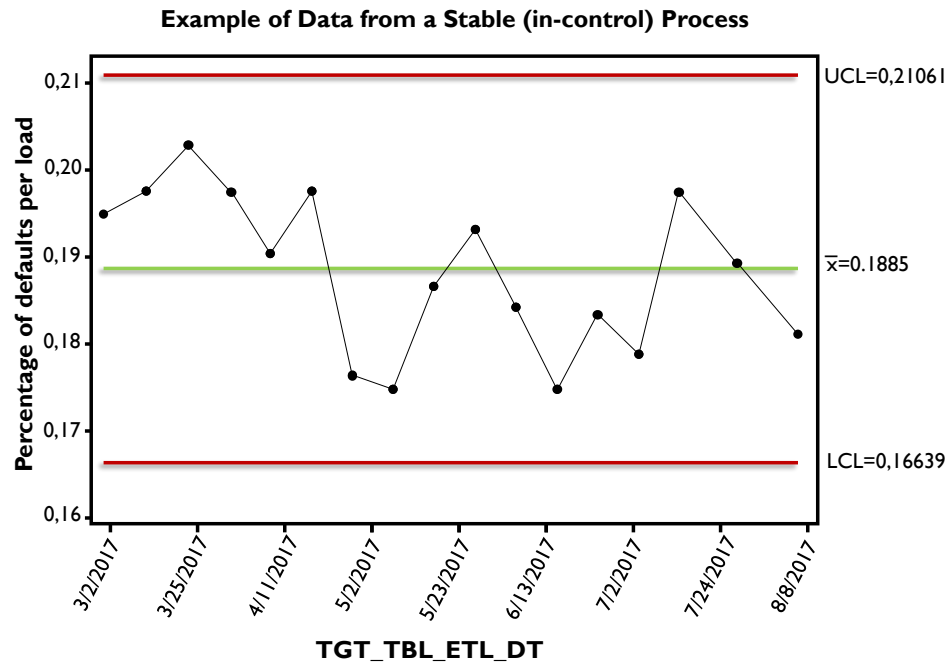


Figure 95 Control Chart of a Process in Statistical Control

SPC measures the predictability of process outcomes by identifying variation within a process. Processes have variation of two types: Common Causes that are inherent in the process and Special Causes that are unpredictable or intermittent. When the only sources of variation are common causes, a system is said to be in (statistical) control and a range of normal variation can be established. This is the baseline against which change can be detected.

Applying SPC to data quality measurement is based on the working assumption that, like a manufactured product, data is the product of a process. Sometimes the process that creates data is very simple (e.g., a person fills out a form). Other times, processes are quite complex: a set of algorithms aggregates medical claim data in order to follow trends related to the effectiveness of particular clinical protocols. If such a process has consistent inputs and is executed consistently, it will produce consistent results each time it is run. However, if the inputs or execution change, then so will the outputs. Each of these components can be measured. The measurements can be used to detect special causes. Knowledge of the special causes can be used to mitigate risks associated with data collection or processing.

SPC is used for control, detection, and improvement. The first step is to measure the process in order to identify and eliminate special causes. This activity establishes the control state of the process. Next is to put in place

measurements to detect unexpected variation as soon as it is detectable. Early detection of problems simplifies investigation of their root causes. Measurements of the process can also be used to reduce the unwanted effects of common causes of variation, allowing for increased efficiency.

4.6 Root Cause Analysis

A root cause of a problem is a factor that, if eliminated, would remove the problem itself. Root cause analysis is a process of understanding factors that contribute to problems and the ways they contribute. Its purpose is to identify underlying conditions that, if eliminated, would mean problems would disappear.

A data management example may clarify the definition. Let's say a data process that runs each month requires as input a file of customer information. Measurement of the data shows that in April, July, October, and January, the quality of the data goes down. Inspection of the timing of delivery shows that in March, June, September, and December, the file is delivered on the 30th of the month, whereas at other times it is delivered on the 25th. Further analysis shows that the team responsible for delivering the file is also responsible for closing quarterly financial processes. These processes take precedence over other work and the files are delivered late during those months, impacting the quality. The root cause of the data quality problem turns out to be a process delay caused by a competing priority. It can be addressed by scheduling file delivery and ensuring that resources can deliver within the schedule.

Common techniques for root cause analysis include Pareto analysis (the 80/20 rule), fishbone diagram analysis, track and trace, process analysis, and the Five Whys (McGilvray, 2008).

5. Implementation Guidelines

Improving the quality of data within an organization is not an easy task – even when data quality improvement efforts are launched from within a data governance program and with the support of senior management. A classic academic discussion is whether it is better to implement a Data Quality program top-down or bottom-up. Typically, a hybrid approach works best – top-down for sponsorship, consistency, and resources, but bottom-up to discover what is actually broken and to achieve incremental successes.

Improving data quality requires changes in how people think about and behave toward data. Cultural change is challenging. It requires planning, training, and reinforcement. (See Chapter 17.) While the specifics of cultural change will differ from organization to organization, most Data Quality program implementations need to plan for:

- **Metrics on the value of data and the cost of poor quality data:** One way to raise organizational awareness of the need for Data Quality Management is through metrics that describe the value of data and the return on investment from improvements. These metrics (which differ from data quality scores) provide the basis for funding improvements and changing the behavior of both staff and management. (See Chapter 11.)

- **Operating model for IT/Business interactions:** Business people know what the important data is, and what it means. Data Custodians from IT understand where and how the data is stored, and so they are well placed to translate definitions of data quality into queries or code that identify specific records that do not comply. (See Chapter 11.)
- **Changes in how projects are executed:** Project oversight must ensure project funding includes steps related to data quality (e.g., profiling and assessment, definition of quality expectations, data issue remediation, prevention and correction, building controls and measurements). It is prudent to make sure issues are identified early and to build data quality expectations upfront in projects.
- **Changes to business processes:** Improving data quality depends on improving the processes by which data is produced. The Data Quality team needs to be able to assess and recommend changes to non-technical (as well as technical) processes that impact the quality of data.
- **Funding for remediation and improvement projects:** Some organizations do not plan for remediating data, even when they are aware of data quality issues. Data will not fix itself. The costs and benefits of remediation and improvement projects should be documented so that work on improving data can be prioritized.
- **Funding for Data Quality Operations:** Sustaining data quality requires ongoing operations to monitor data quality, report on findings, and continue to manage issues as they are discovered.

5.1 Readiness Assessment / Risk Assessment

Most organizations that depend on data have a lot of opportunity for improvement. How formal and well-supported a Data Quality program will be depends on how mature the organization is from a data management perspective. (See Chapter 15.) Organizational readiness to adopt data quality practices can be assessed by considering the following characteristics:

- **Management commitment to managing data as a strategic asset:** As part of asking for support for a Data Quality program, it is important to determine how well senior management understands the role that data plays in the organization. To what degree does senior management recognize the value of data to strategic goals? What risks do they associate with poor quality data? How knowledgeable are they about the benefits of data governance? How optimistic about the ability to change culture to support quality improvement?
- **The organization's current understanding of the quality of its data:** Before most organizations start their quality improvement journey, they generally understand the obstacles and pain points that signify poor quality data. Gaining knowledge of these is important. Through them, poor quality data can be directly associated with negative effects, including direct and indirect costs, on the organization. An understanding of pain points also helps identify and prioritize improvement projects.
- **The actual state of the data:** Finding an objective way to describe the condition of data that is causing pain points is the first step to improving the data. Data can be measured and described through

profiling and analysis, as well as through quantification of known issues and pain points. If the DQ team does not know the actual state of the data, then it will be difficult to prioritize and act on opportunities for improvement.

- **Risks associated with data creation, processing, or use:** Identifying what can go wrong with data and the potential damage to an organization from poor quality data provides the basis for mitigating risks. If the organization does not recognize these risks, it may be challenging to get support for the Data Quality program.
- **Cultural and technical readiness for scalable data quality monitoring:** The quality of data can be negatively impacted by business and technical processes. Improving the quality of data depends on cooperation between business and IT teams. If the relationship between business and IT teams is not collaborative, then it will be difficult to make progress.

Findings from a readiness assessment will help determine where to start and how quickly to proceed. Findings can also provide the basis for road-mapping program goals. If there is strong support for data quality improvement and the organization knows its own data, then it may be possible to launch a full strategic program. If the organization does not know the actual state of its data, then it may be necessary to focus on building that knowledge before developing a full strategy.

5.2 Organization and Cultural Change

The quality of data will not be improved through a collection of tools and concepts, but through a mindset that helps employees and stakeholders to act while always thinking of the quality of data and what the business and their customers need. Getting an organization to be conscientious about data quality often requires significant cultural change. Such change requires vision and leadership. (See Chapter 17.)

The first step is promoting awareness about the role and importance of data to the organization. All employees must act responsibly and raise data quality issues, ask for good quality data as consumers, and provide quality information to others. Every person who touches the data can impact the quality of that data. Data quality is not just the responsibility of a DQ team or IT group.

Just as the employees need to understand the cost to acquire a new customer or retain an existing customer, they also need to know the organizational costs of poor quality data, as well as the conditions that cause data to be of poor quality. For example, if customer data is incomplete, a customer may receive the wrong product, creating direct and indirect costs to an organization. Not only will the customer return the product, but he or she may call and complain, using call center time, with the potential for reputational damage to the organization. If customer data is incomplete because the organization has not established clear requirements, then everyone who uses this data has a stake in clarifying requirements and following standards.

Ultimately, employees need to think and act differently if they are to produce better quality data and manage data in ways that ensures quality. This requires training and reinforcement. Training should focus on:

- Common causes of data problems

- Relationships within the organization's data ecosystem and why improving data quality requires an enterprise approach
- Consequences of poor quality data
- Necessity for ongoing improvement (why improvement is not a one-time thing)
- Becoming 'data-lingual', about to articulate the impact of data on organizational strategy and success, regulatory reporting, customer satisfaction

Training should also include an introduction to any process changes, with assertions about how the changes improve data quality.

6. Data Quality and Data Governance

A Data Quality program is more effective when part of a data governance program. Often data quality issues are the reason for establishing enterprise-wide data governance (see Chapter 3). Incorporating data quality efforts into the overall governance effort enables the Data Quality program team to work with a range of stakeholders and enablers:

- Risk and security personnel who can help identify data-related organizational vulnerabilities
- Business process engineering and training staff who can help teams implement process improvements
- Business and operational data stewards, and data owners who can identify critical data, define standards and quality expectations, and prioritize remediation of data issues

A Governance Organization can accelerate the work of a Data Quality program by:

- Setting priorities
- Identifying and coordinating access to those who should be involved in various data quality-related decisions and activities
- Developing and maintaining standards for data quality
- Reporting relevant measurements of enterprise-wide data quality
- Providing guidance that facilitates staff involvement
- Establishing communications mechanisms for knowledge-sharing
- Developing and applying data quality and compliance policies
- Monitoring and reporting on performance
- Sharing data quality inspection results to build awareness, identify opportunities for improvements, and build consensus for improvements
- Resolving variations and conflicts; providing direction

6.1 Data Quality Policy

Data Quality efforts should be supported by and should support data governance policies. For example, governance policies can authorize periodic quality audits and mandate compliance to standards and best

practices. All Data Management Knowledge Areas require some level of policy, but data quality policies are particularly important as they often touch on regulatory requirements. Each policy should include:

- Purpose, scope and applicability of the policy
- Definitions of terms
- Responsibilities of the Data Quality program
- Responsibilities of other stakeholders
- Reporting
- Implementation of the policy, including links to risk, preventative measures, compliance, data protection, and data security

6.2 Metrics

Much of the work of a Data Quality team will focus on measuring and reporting on quality. High-level categories of data quality metrics include:

- **Return on Investment:** Statements on cost of improvement efforts vs. the benefits of improved data quality
- **Levels of quality:** Measurements of the number and percentage of errors or requirement violations within a data set or across data sets
- **Data Quality trends:** Quality improvement over time (i.e., a trend) against thresholds and targets, or quality incidents per period
- **Data issue management metrics:**
 - Counts of issues by dimensions of data quality
 - Issues per business function and their statuses (resolved, outstanding, escalated)
 - Issue by priority and severity
 - Time to resolve issues
- **Conformance to service levels:** Organizational units involved and responsible staff, project interventions for data quality assessments, overall process conformance
- **Data Quality plan rollout:** As-is and roadmap for expansion

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Big Data and Data Science

1. Introduction

Since the early 2000s, the terms *Big Data* and *Data Science* have, unfortunately, been bandied about as buzzwords. The concepts and their implications are misunderstood – or, at least, there is limited consensus on their meaning. Even the meaning of ‘Big’ is relative. That said, both Big Data and Data Science are connected to significant technological changes that have allowed people to generate, store, and analyze larger and larger amounts of data. More importantly, people can use that data to predict and influence behavior, as well as to gain insight on a range of important subjects, such as health care practices, natural resource management, and economic development.

Big Data refers not only to the volume of data, but also to its variety (structured and unstructured, documents, files, audio, video, and streaming data, etc.), and the speed at which it is produced (velocity). People who mine and develop predictive, machine learning, and prescriptive models and analytics from these and deploy results for analysis by interested parties are called Data Scientists.

Data Science has existed for a long time; it used to be called ‘applied statistics’. But the capability to explore data patterns has quickly evolved in the twenty-first century with the advent of Big Data and the technologies that support it. Traditional Business Intelligence provides ‘rear-view mirror’ reporting – analysis of structured data to describe past trends. In some cases, BI patterns are used to predict future behavior, but not with high confidence. Until recently, in-depth analysis of enormous data sets has been limited by technology. Analyses have relied on sampling or other means of abstraction to approximate patterns. As the capacity to collect and analyze large data sets has grown, Data Scientists have integrated methods from mathematics, statistics, computer science, signal processing, probability modeling, pattern recognition, machine learning, uncertainty modeling, and data visualization in order to gain insight and predict behaviors based on Big Data sets. In short, Data Science has found new ways to analyze and get value from data.

As Big Data has been brought into data warehousing and Business Intelligence environments, Data Science techniques are used to provide a forward-looking (‘windshield’) view of the organization. Predictive capabilities, real-time and model-based, using different types of data sources, offer organizations better insight into where they are heading. (See Figure 96.)

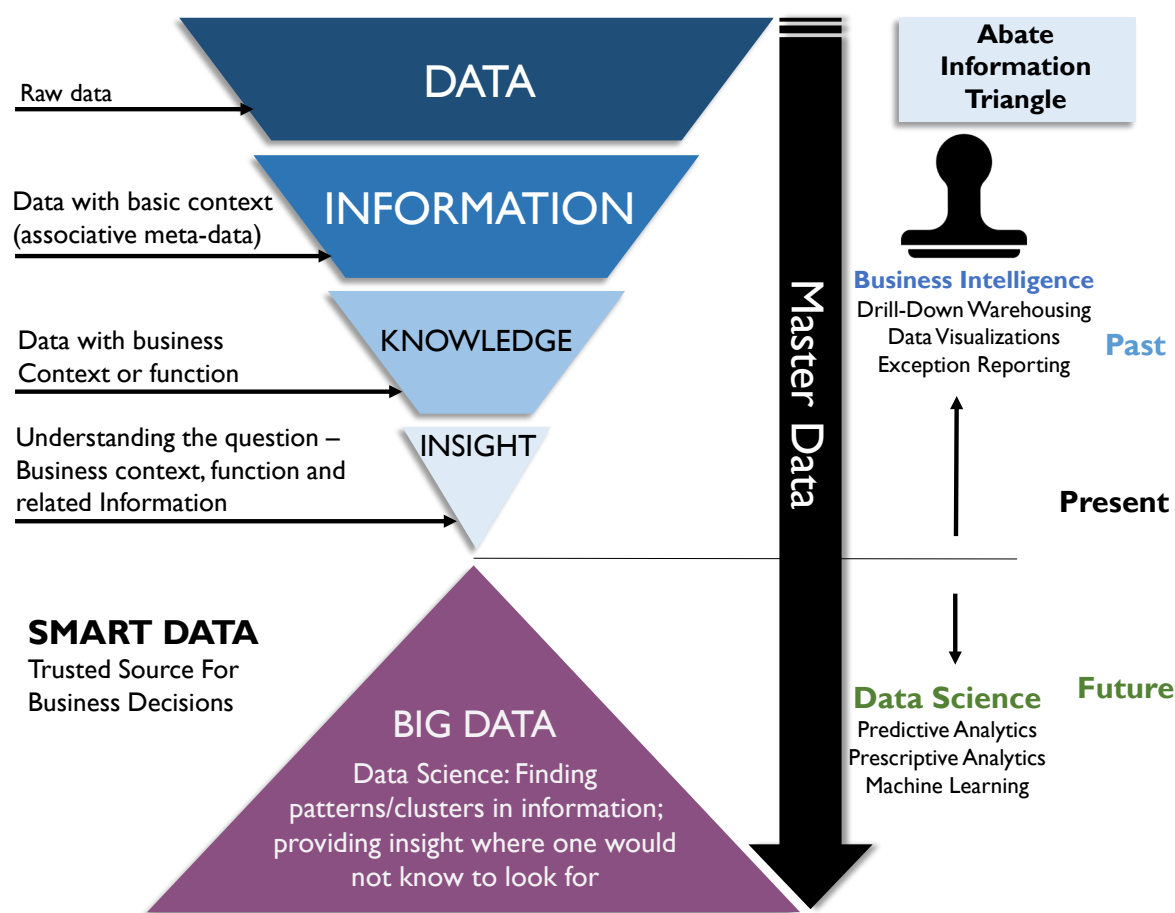


Figure 96 Abate Information Triangle

To take advantage of Big Data, however, requires change in the way that data is managed. Most data warehouses are based on relational models. Big Data is not generally organized in a relational model. Most data warehousing depends on the concept of ETL (Extract, Transform, and Load). Big Data solutions, like data lakes, depend on the concept of ELT – loading and then transforming. As importantly, the speed and volume of data present challenges that require different approaches to critical aspects of data management, such as integration, Metadata Management, and Data Quality assessment.

1.1 Business Drivers

The biggest business driver for developing organizational capabilities around Big Data and Data Science is the desire to find and act on business opportunities that may be discovered through data sets generated through a diversified range of processes. Big Data can stimulate innovation by making more and larger data sets available for exploration. This data can be used to define predictive models that anticipate customer needs and enable personalized presentation of products and services. Data Science can improve operations. Machine learning algorithms can automate complex time-consuming activities, thus improving organizational efficiency, reducing costs, and mitigating risks.

Big Data and Data Science

Definition: The collection (Big Data) and analysis (Data Science, Analytics and Visualization) of many different types of data to find answers and insights for questions that are not known at the start of analysis.

Goals:

1. Discover relationships between data and the business.
2. Support the iterative integration of data source(s) into the enterprise.
3. Discover and analyze new factors that might affect the business.
4. Publish data using visualization techniques in an appropriate, trusted, and ethical manner.

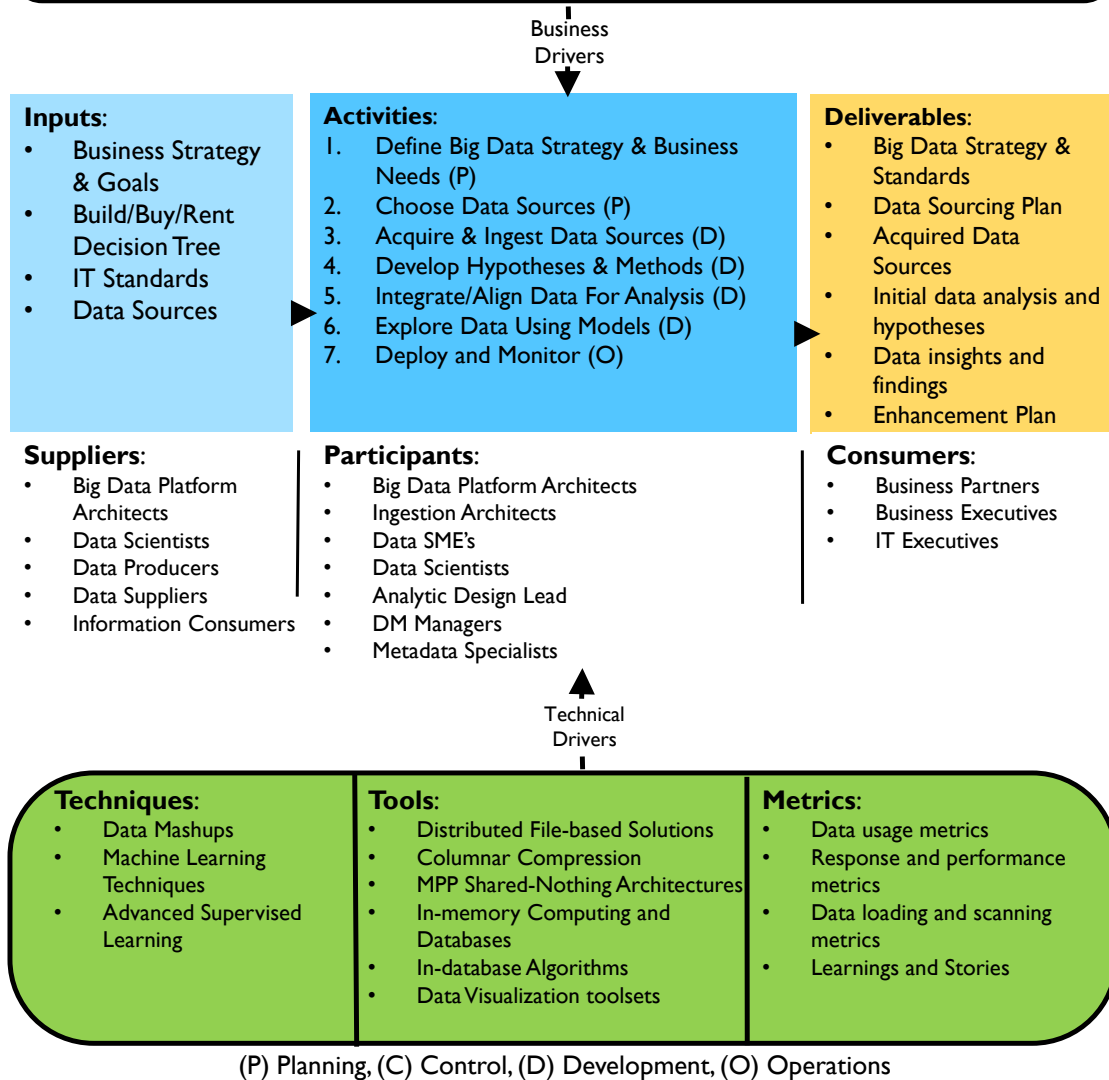


Figure 97 Context Diagram: Big Data and Data Science

1.2 Principles

The promise of Big Data – that it will provide a different kind of insight – depends on being able to manage Big Data. In many ways, because of the wide variation in sources and formats, Big Data management will require more discipline than relational data management. Principles related to Big Data management have yet to fully form, but one is very clear: Organizations should carefully manage Metadata related to Big Data sources in order to have an accurate inventory of data files, their origins, and their value.

1.3 Essential Concepts

1.3.1 Data Science

As noted in the chapter introduction, Data Science merges data mining, statistical analysis, and machine learning with data integration and data modeling capabilities, to build predictive models that explore data content patterns. Developing predictive models is sometimes called *Data Science* because the data analyst, or data scientist, uses the scientific method to develop and assess a model.

The data scientist develops a hypothesis about behavior that can be observed in the data prior to a particular action. For example, the purchase of one type of item is usually followed by the purchase of another type of item (the purchase of a house is usually followed by the purchase of furniture). Then, the data scientist analyzes large amounts of historical data to determine how frequently the hypothesis has been true in the past and to statistically verify the probable accuracy of the model. If a hypothesis is valid with sufficient frequency, and if the behavior it predicts is useful, then the model may become the basis for an operational intelligence process to predict future behavior, even possibly in real time such as suggestive selling advertisements.

Developing Data Science solutions involves the iterative inclusion of data sources into models that develop insights. Data Science depends on:

- **Rich data sources:** Data with the potential to show otherwise invisible patterns in organizational or customer behavior
- **Information alignment and analysis:** Techniques to understand data content and combine data sets to hypothesize and test meaningful patterns
- **Information delivery:** Running models and mathematical algorithms against the data and producing visualizations and other output to gain insight into behavior
- **Presentation of findings and data insights:** Analysis and presentation of findings so that insights can be shared

Table 32 compares the role of traditional DW/BI to predictive and prescriptive analytics that can be achieved through Data Science techniques.

Table 32 Analytics Progression

DW / Traditional BI	Data Science	
Descriptive	Predictive	Prescriptive
Hindsight	Insight	Foresight
Based on history: What happened? Why did it happen?	Based on predictive models: What is likely to happen?	Based on scenarios: What should we do to make things happen?

1.3.2 The Data Science Process

Figure 98 illustrates the iterative phases of the Data Science process. The outputs of each step become the inputs into the next. (See Section 2.)

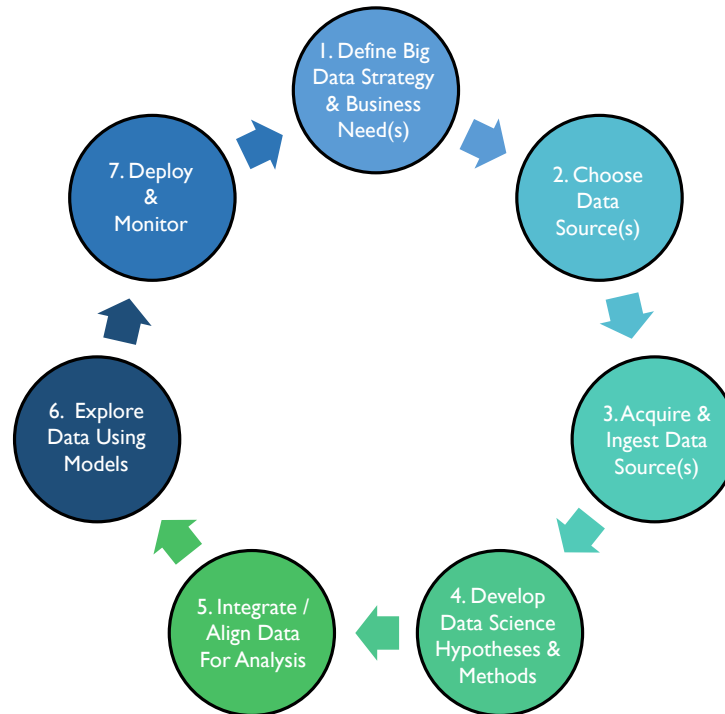


Figure 98 Data Science Process

The Data Science process follows the scientific method of refining knowledge by making observations, formulating and testing hypotheses, observing results, and formulating general theories that explain results. Within Data Science, this process takes the form of observing data and creating and evaluating models of behavior:

- **Define Big Data strategy and business needs:** Define the requirements that identify desired outcomes with measurable tangible benefits.
- **Choose data sources:** Identify gaps in the current data asset base and find data sources to fill those gaps.

- **Acquire and ingest data sources:** Obtain data sets and onboard them.
- **Develop Data Science hypotheses and methods:** Explore data sources via profiling, visualization, mining, etc.; refine requirements. Define model algorithm inputs, types, or model hypotheses and methods of analysis (i.e., groupings of data found by clustering, etc.).
- **Integrate and align data for analysis:** Model feasibility depends in part on the quality of the source data. Leverage trusted and credible sources. Apply appropriate data integration and cleansing techniques to increase quality and usefulness of provisioned data sets.
- **Explore data using models:** Apply statistical analysis and machine learning algorithms against the integrated data. Validate, train, and over time, evolve the model. Training entails repeated runs of the model against actual data to verify assumptions and make adjustments, such as identifying outliers. Through this process, requirements will be refined. Initial feasibility metrics guide evolution of the model. New hypotheses may be introduced that require additional data sets and results of this exploration will shape the future modeling and outputs (even changing the requirements).
- **Deploy and monitor:** Those models that produce useful information can be deployed to production for ongoing monitoring of value and effectiveness. Often Data Science projects turn into data warehousing projects where more vigorous development processes are put in place (ETL, DQ, Master Data, etc.).

1.3.3 Big Data

Early efforts to define the meaning of Big Data characterized it in terms of the Three V's: Volume, Velocity, Variety (Laney, 2001). As more organizations start to leverage the potential of Big Data, the list of V's has expanded:

- **Volume:** Refers to the amount of data. Big Data often has thousands of entities or elements in billions of records.
- **Velocity:** Refers to the speed at which data is captured, generated, or shared. Big Data is often generated and can also be distributed and even analyzed in real-time.
- **Variety / Variability:** Refers to the forms in which data is captured or delivered. Big Data requires storage of multiple formats; data structure is often inconsistent within or across data sets.
- **Viscosity:** Refers to how difficult the data is to use or integrate.
- **Volatility:** Refers to how often data changes occur and therefore how long the data is useful.
- **Veracity:** Refers to how trustworthy the data is.

Big Data volumes are exceptionally large (greater than 100 Terabyte and often in the Petabyte and Exabyte range). In warehousing and analytic solutions, very large volumes of data pose challenges to data loading, modeling, cleansing, and analytics. These challenges are often solved using massively parallel processing, or

parallel processing and distributed data solutions. However, they have much wider implications. The size of data sets requires changes in the overall way that data is stored and accessed, as well as in how data is understood (e.g., much of our current way of thinking about data is based on relational database structures), as well as how data is managed (Adams, 2009). Figure 99 presents a visual summary of the range of data that has become available through Big Data technologies and the implications on data storage options.

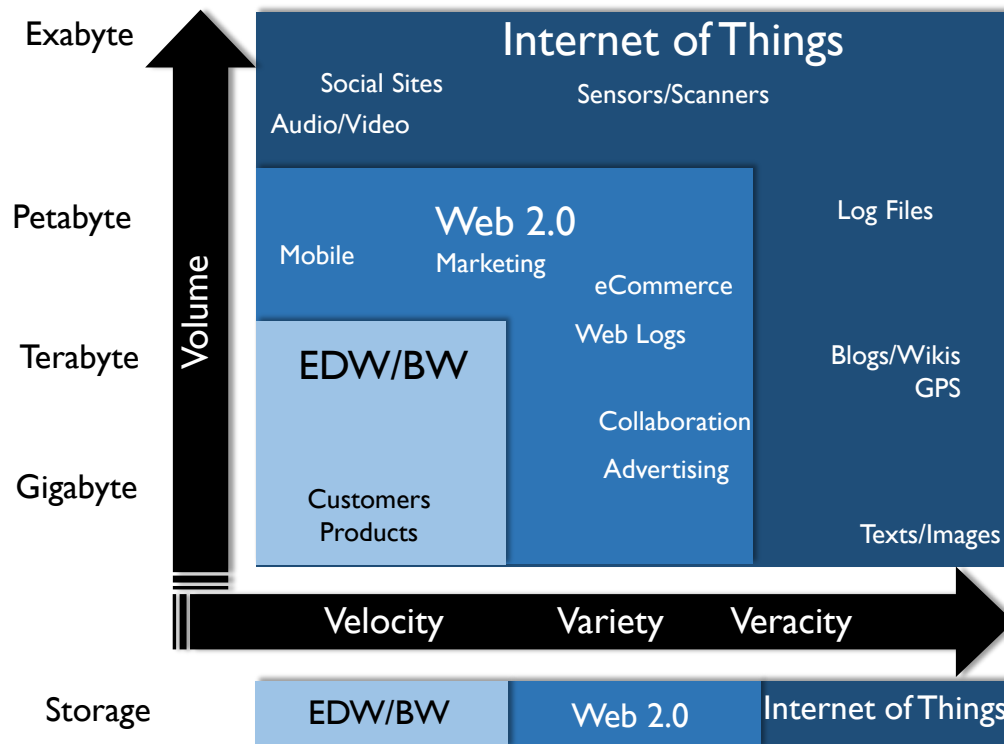


Figure 99 Data Storage Challenges⁸⁸

1.3.4 Big Data Architecture Components

The selection, installation, and configuration of a Big Data and Data Science environment require specialized expertise. End-to-end architectures must be developed and rationalized against existing data exploratory tools and new acquisitions.

Figure 100 describes the DW/BI and Big Data Architecture. (Details about DW/BI components are described in Chapter 11.) The biggest difference between DW/BI and Big Data processing is that in a traditional data warehouse, data is integrated as it is brought into the warehouse (extract, TRANSFORM, load); while in a Big Data environment, data is ingested and loaded before it is integrated (extract, LOAD, transform). In some cases, data may not be integrated at all, in the traditional sense. Instead of being integrated in preparation for use, it is often integrated through particular uses (e.g., the process of building predictive models drives the integration of particular data sets).

⁸⁸ Sourced and used with permission from Robert Abate / EMC Corporation.

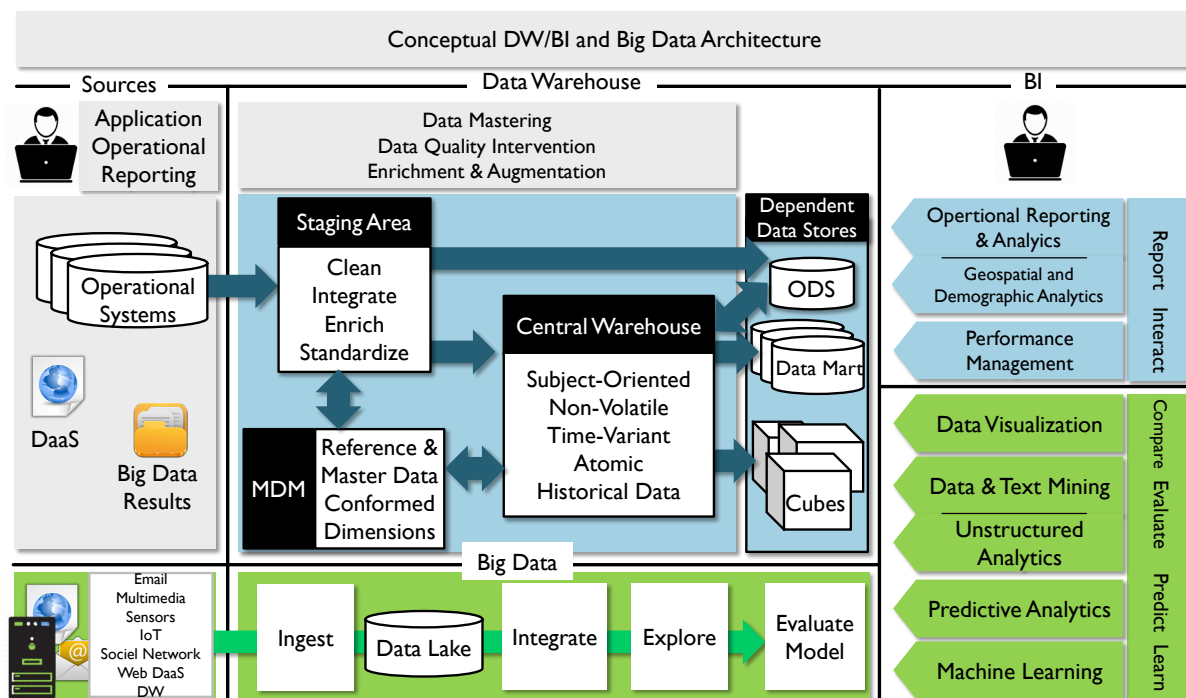


Figure 100 Conceptual DW/BI and Big Data Architecture

The difference between ETL and ELT has significant implications for how data is managed. For example, the integration process does not necessarily rely on or produce an enterprise data model. The risk is that much knowledge about data can be lost if processes for ingestion and use are executed in an ad hoc way. There is a need to collect and manage Metadata related to these processes, if they are to be understood and leveraged overtime.

The following sections will describe Big Data Sources and the Data Lake construct. The activities (Ingest, Integrate, Explore, Evaluate Model) are explored under the Activities section.

1.3.5 Sources of Big Data

Because so much of human activity is executed electronically, massive amounts of data are accumulating every day as we move through the world, interact with each other, and transact business. Big Data is produced through email, social media, online orders, and even online video games. Data is generated not only by phones and point-of-sale devices, but also by surveillance systems, sensors in transportation systems, medical monitoring systems, industrial and utility monitoring systems, satellites, and military equipment. For example, one airline flight can generate a terabyte of data. Devices that interact directly with the Internet generate a large portion of Big Data. The connections between devices and the Internet are sometimes called the Internet of Things (IoT).

1.3.6 Data Lake

A data lake is an environment where a vast amount of data of various types and structures can be ingested, stored, assessed, and analyzed. Data lakes can serve many purposes. For example, providing

- An environment for Data Scientists to mine and analyze data
- A central storage area for raw data, with minimal, if any, transformation
- Alternate storage for detailed historical data warehouse data
- An online archive for records
- An environment to ingest streaming data with automated pattern identification

A data lake can be implemented as a complex configuration of data handling tools including Hadoop or other data storage systems, cluster services, data transformation, and data integration. These handlers have facilitated cross-infrastructure, analytic facilitation software to bring the configuration together.

The risk of a data lake is that it can quickly become a data swamp – messy, unclear, and inconsistent. In order to establish an inventory of what is in a data lake, it is critical to manage Metadata as the data is ingested. In order to understand how the data in a data lake is associated or connected, data architects or data engineers often use unique keys or other techniques (semantic models, data models, etc.) so that data scientists and other visualization developers know how to use the information stored within the data lake. (See Chapter 9.)

1.3.7 Services-Based Architecture

Services-based architecture (SBA) is emerging as a way to provide immediate (if not completely accurate or complete) data, as well as update a complete, accurate historical data set, using the same source (Abate, Aiken, Burke, 1997). The SBA architecture is similar to the DW architectures which send data directly to an ODS for immediate access, as well as to the DW for historical accumulation. SBA architectures have three main components, a batch layer, a speed layer, and a serving layer. (See Figure 101.)

- **Batch layer:** A data lake serves as the batch layer, containing both recent and historical data
- **Speed layer:** Contains only real-time data
- **Serving layer:** Provides an interface to join data from the batch and speed layers

Data is loaded into both the batch and speed layers. All analytic computations are performed on data in both the batch and speed layers, which most likely requires implementation in two separate systems. Organizations address synchronization issues through trade-offs between completeness, latency, and complexity of merged views defined in the serving layer. Cost/benefit assessment is required to determine whether reducing latency or improving data completeness is worth the associated cost and complexity.

The batch layer is often referred to as the structure-over-time component (here every transaction is an insert), whereas in the speed layer (often referred to as an Operational Data Store or ODS), all transactions are updates (or inserts only if required). In this manner, the architecture prevents synchronization issues while simultaneously creating a current state and a history layer. This architecture usually provides its data through a

serving or data services layer that abstracts the data utilizing Metadata. This services layer determines where the data is to be ‘served’ from and appropriately provides the data requested.

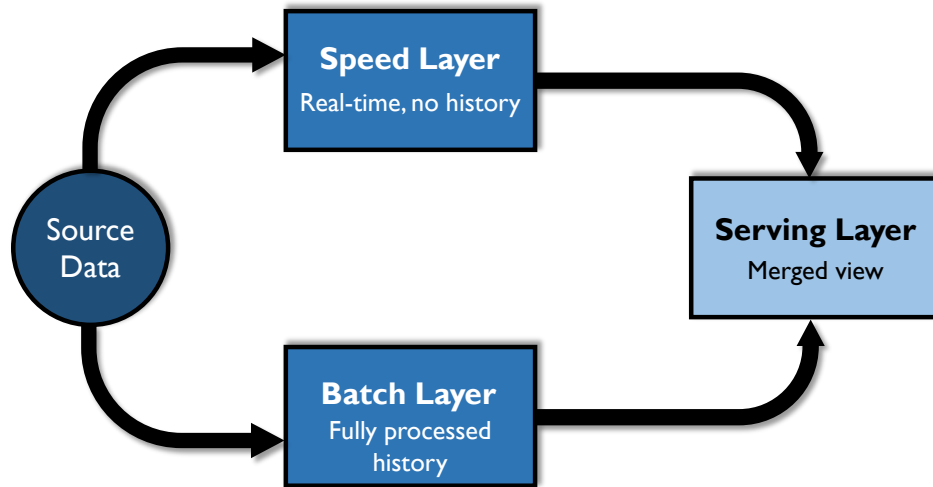


Figure 101 Services-based Architecture

1.3.8 Machine Learning

Machine Learning explores the construction and study of learning algorithms. It can be viewed as a union of unsupervised learning methods, more commonly referred to as data mining, and supervised learning methods deeply rooted in mathematical theory, specifically statistics, combinatorics, and optimization. A third branch is now forming called reinforcement learning, where goal performance is earned but not specifically teacher recognized – driving a vehicle for example. Programming machines to quickly learn from queries and adapt to changing data sets led to a completely new field within Big Data referred to as machine learning.⁸⁹ Processes run, and results are stored that are then used in subsequent runs to iteratively inform the process and refine the results.

Machine Learning explores the construction and study of learning algorithms. These algorithms fall into three types:

- **Supervised learning:** Based on generalized rules; for example, separating SPAM from non-SPAM email
- **Unsupervised learning:** Based on identifying hidden patterns (i.e., data mining)
- **Reinforcement learning:** Based on achieving a goal (e.g., beating an opponent at chess)

Statistical modeling and machine learning have been employed to automate otherwise costly research and development projects, by performing several trial and error passes on a vast set of data, repeating trials with the results collected, analyzed, and errors corrected. This approach can reduce time to answer dramatically and

⁸⁹ Refer to the Machine Learning resources periodic table at <http://bit.ly/1DpTrHC> for an interactive guide to the differing platforms available for the machine learning developer, scientist, and practitioner.

guide organizational initiatives with insights based on cost effective repeatable processes. For example, CIVDDD uses machine learning and complex scientific data visualization techniques to help government agencies and peacekeepers meet the challenge of dealing with the masses of threat-related information.⁹⁰

While it taps into data in new ways, machine learning has ethical implications, especially with respect to the principle of transparency. Evidence shows that deep learning neural networks (DLNN) work. They learn things. However, it is not always clear how they learn. As the algorithms that drive these processes become more complex, they also become more opaque, functioning as ‘black boxes’. As they account for a greater number of variables and as those variables themselves are more abstract, algorithms test the limits of human ability to interpret the machine (Davenport, 2017). The need for transparency – the ability to see how decisions are made – will likely increase as this functionality evolves and is put to use in a wider array of situations. (See Chapter 2.)

1.3.9 Sentiment Analysis

Media monitoring and text analysis are automated methods for retrieving insights from large unstructured or semi-structured data, such as transaction data, social media, blogs, and web news sites. This is used to understand what people say and feel about brands, products, or services, or other types of topics. Using Natural Language Processing (NLP) or by parsing phrases or sentences, semantic analysis can detect sentiment and also reveal changes in sentiment to predict possible scenarios.

Consider the case of looking for key words in a posting. If the words *good* or *great* are present, this might be a positive response, versus seeing *awful* or *bad* might be signs that this could be a negative response. Categorizing the data into the types of responses, the ‘sentiment’ of the whole community or posting (social media such as Twitter, blogs, etc.) is exposed. That said, sentiment is not easily gained, as the words by themselves do not tell the whole story (i.e., I had a *Great* problem with their customer service). Sentiment must interpret words in context. This requires an understanding of the meaning of the post – this interpretation often requires work using NLP functions found in such systems as IBM’s Watson.

1.3.10 Data and Text Mining

Data mining is a particular kind of analysis that reveals patterns in data using various algorithms. It began as an offshoot from Machine Learning, a subfield of Artificial Intelligence. The theory is a subset of statistical analysis known as unsupervised learning where algorithms are applied to a data set without knowledge or intent of the desired outcome. While standard query and reporting tools ask specific questions, data mining tools help discover unknown relationships by revealing patterns. Data mining is a key activity during the exploration phase as it facilitates rapid identification of studied data elements, identifies new relationships previously unknown, unclear, or unclassified, and provides structure for the classification of studied data elements.

⁹⁰ CIVDDD, the Centre for Innovation in Information and Data-Driven Design, is a research grant in big data analytics and visualization to develop next generation data discovery, design, and visualization techniques for new computational tools, representational strategies, and interfaces.

Text mining analyzes documents with text analysis and data mining techniques to classify content automatically into workflow-guided and SME-directed ontologies. Thus, electronic text media can be analyzed without restructuring or reformatting. Ontologies can be linked into search engines, allowing for web-enabled querying against these documents. (See Chapter 9.)

Data and text mining use a range of techniques, including:

- **Profiling:** Profiling attempts to characterize the typical behavior of an individual, group, or population. Profiling is used to establish behavioral norms for anomaly detection applications, such as fraud detection and monitoring for intrusions to computer systems. Profile results are inputs for many unsupervised learning components.
- **Data reduction:** Data reduction replaces a large data set with a smaller set of data that contains much of the important information in the larger set. The smaller data set may be easier to analyze or process.
- **Association:** Association is an unsupervised learning process to find relationships between studied elements based on transactions involving them. Examples of association include: Frequent item set mining, rule discovery, and market-based analysis. Recommendation systems on the internet use this process as well.
- **Clustering:** Clustering group elements in a study together by their shared characteristics. Customer segmentation is an example of clustering.
- **Self-organizing maps:** Self-organizing maps are a neural network method of cluster analysis. Sometimes referred to as Kohonen Maps, or topologically ordered maps, they aim to reduce the dimensionality in the evaluation space while preserving distance and proximity relationships as much as possible, akin to multi-dimension scaling. Reducing the dimensionality is like removing one variable from the equation without violating the outcome. This makes it easier to solve and visualize.

1.3.11 Predictive Analytics

Predictive Analytics is the sub-field of supervised learning where users attempt to model data elements and predict future outcomes through evaluation of probability estimates. Rooted deeply in mathematics specifically statistics, predictive analytics shares many components with unsupervised learning, with the prescribed difference to the measurement of a desired predictive outcome.

Predictive Analytics is the development of probability models based on variables, including historical data, related to possible events (purchases, changes in price, etc.). When it receives other pieces of information, the model triggers a reaction by the organization. The triggering factor may be an event, such as a customer adding a product to an on-line shopping basket, or it may be data in a data stream, such as a news feed or utility sensor data, or an increased volume of service requests. The triggering factor may be an external event. News being reported about a company is a big predictor of a change in stock price. Predicting stock movement should include monitoring news and determining if news about a company is likely to be good or bad for the stock price.

Frequently, the triggering factor is the accumulation of a large volume of real-time data, such as an extremely high number of trades or requests for service or volatility of the environment. Monitoring a data event stream includes incrementally building on the populated models until a threshold is reached as defined in the model.

The amount of time that a predictive model provides between the prediction and event predicted is frequently very small (seconds or less than a second). Investment in very low latency technology solutions, such as in-memory databases, high-speed networks, and even physical proximity to the source of the data, optimizes an organization's ability to react to the prediction.

The simplest form of predictive model is the forecast. Many techniques exist for trending or forecasting based on regression analysis and benefit from smoothing. The simplest way to smooth data is through a moving average, or even a weighted moving average. More advanced techniques can be useful, like the exponential moving average, which introduces a smoothing factor to be applied. Minimizing the error residual from the least squares can be a starting point, but several runs are necessary to determine and optimize the smoothing factor. Double and triple exponential smoothing models exist to address trend and seasonality components.

1.3.12 Prescriptive Analytics

Prescriptive analytics take predictive analytics a step farther to define actions that will affect outcomes, rather than just predicting the outcomes from actions that have occurred. Prescriptive analytics anticipates what will happen, when it will happen, and implies why it will happen. Because prescriptive analytics can show the implications of various decisions, it can suggest how to take advantage of an opportunity or avoid a risk. Prescriptive analytics can continually take in new data to re-predict and re-prescribe. This process can improve prediction accuracy and result in better prescriptions.

1.3.13 Unstructured Data Analytics

Unstructured data analytics combines text mining, association, clustering, and other unsupervised learning techniques to codify large data sets. Supervised learning techniques can also be applied to provide orientation, oversight, and guidance in the coding process leveraging human intervention to resolve ambiguity when necessary.

Unstructured data analytics is becoming more important as more unstructured data is generated. Some analysis is impossible without the ability to incorporate unstructured data into analytical models. However, unstructured data is difficult to analyze without some way to isolate the elements of interest from extraneous elements.

Scanning and tagging is one way to add 'hooks' to unstructured data that allow filtering and linking to related structured data. However, knowing what tags to generate based on what conditions is difficult. It is an iterative process, from when proposed tag conditions are identified, tags are assigned as data is ingested, then analytics uses those tags to validate the tag condition, and analyze the tagged data, which then leads to potentially changed tag conditions, or more tags.

1.3.14 Operational Analytics

The concept of operational analytics (also known as operational BI or streaming analytics) has emerged from the integration of real-time analytics into operations. Operational analytics includes activities like user segmentation, sentiment analysis, geocoding, and other techniques applied to data sets for marketing campaign analysis, sales penetration, product adoption, asset optimization, and risk management.

Operational analytics involves tracking and integrating real-time streams of information, deriving conclusions based on predictive models of behavior, and triggering automatic responses and alerts. Designing the model, triggers, and responses required for successful analysis takes more analysis on the data itself. An operational analytics solution includes the preparation of historical data for pre-population of the models of behavior. For example, in a retail product model, populating a shopping basket analysis that identifies products often purchased together. In predicting behavior of financial markets, historical price information and historical price rate of change are regularly used. Pre-population calculations are usually performed in advance to enable timely responses to triggering events.

Once predictive models have been determined to be both useful and cost effective, solutions which integrate historical and current data (including real-time and streaming data, structured and unstructured) are implemented to populate the predictive models and trigger actions based on the predictions. The solution must ensure real-time data streams using the model rules are processed correctly and automated responses to meaningful events in the data are generated correctly.

1.3.15 Data Visualization⁹¹

Visualization is the process of interpreting concepts, ideas, and facts by using pictures or graphical representations. Data visualization facilitates understanding of the underlying data by presenting it in a visual summary, such as a chart or graph. Data visualizations condense and encapsulate characteristics data, making them easier to see. In doing so, they can surface opportunities, identify risks, or highlight messages.

Data visualizations can be delivered in a static format, such as a published report, or a more interactive on-line format; and some support end-user interaction where drilling or filtering capabilities facilitate analysis of data within the visualization. Others allow the visualization to be changed by the user on demand through innovative displays, such as data maps and moving landscapes of data over time.

Visualization has long been critical to data analysis. Traditional BI tools include visualization options such as tables, pie charts, lines charts, area charts, bar charts, histograms, and turnkey boxes (candlesticks). To meet the growing need to understand data, the number of visualization tools has increased and techniques have improved.

⁹¹ Data visualization is an evolving field. Principles applied in data visualization are based on design principles. See Tufte, 2001 and McCandless 2012. Numerous web-based resources exist with examples and counter examples. See the Periodic Table of Visualization Methods at Visual Literacy.Org <http://bit.ly/IX1bvI>.

As data analytics matures, visualizing data in new ways will offer strategic advantages. Seeing new patterns in data can result in new business opportunities. As data visualization continues to evolve, organizations will have to grow their Business Intelligence teams to compete in an increasingly data-driven world. Business analytical departments will seek data experts with visualization skills, including data scientists, data artists, and data vision experts, in addition to traditional information architects and data modelers, especially given the risks associated with misleading visualization. (See Chapter 2.)

1.3.16 Data Mashups

Mashups combine data and services to create visualization for insight or analysis. Many virtualization tools enable mashups through functionality that relates data sources by common data elements, originally used to relate a name or descriptive text to a stored code. This client presentation mashup technique is ideal during discovery or exploration phases as they provide immediate benefits. This technique can be readily applied to the web where secured data mashups enable sharing of personal or confidential information across suppliers or providers. These can couple with artificial intelligence learning algorithms to expose internet-based services with natural language interfaces.

2. Activities

2.1 Define Big Data Strategy and Business Needs

An organization's Big Data strategy needs to be aligned with and support its overall business strategy and business requirements and be part of its data strategy. A Big Data strategy must include criteria to evaluate:

- **What problems the organization is trying to solve. What it needs analytics for:** While one advantage of Data Science is that it can provide a new perspective on an organization, the organization still needs to have a starting point. An organization may determine that the data is to be used to understand the business or the business environment; to prove ideas about the value of new products; to explore something that is unknown; or to invent a new way to do business. It is important to establish a gating process to evaluate these initiatives at several phases during the implementation. The value and feasibility of initiatives need to be evaluated at several points in time.
- **What data sources to use or acquire:** Internal sources may be easy to use, but may also be limited in scope. External sources may be useful, but are outside operational control (managed by others, or not controlled by anyone, as in the case of social media). Many vendors are competing in this space and often multiple sources exist for the desired data elements or sets. Acquiring data that integrates with existing ingestion items can reduce overall investment costs.

- **The timeliness and scope of the data to provision:** Many elements can be provided in real-time feeds, snapshots at a point in time, or even integrated and summarized. Low latency data is ideal, but often comes at the expense of machine learning capabilities – there is a huge difference between computational algorithms directed to data-at-rest versus streaming. Do not minimize the level of integration required for downstream usage.
- **The impact on and relation to other data structures:** There may need to be structure or content changes in other data structures to make them suitable for integration with Big Data sets.
- **Influences to existing modeled data:** Including extending the knowledge on customers, products, and marketing approaches.

The strategy will drive the scope and timing of an organization's Big Data capability roadmap.

2.2 Choose Data Sources

As with any development project, the choice of data sources for Data Science work must be driven by the problems the organization is trying to solve. The difference with Big Data / Data Science development is that the range of data sources is wider. It is not limited by format and can include data both external to and internal to an organization. The ability to incorporate this data into a solution also comes with risks. The quality and reliability of the data needs to be evaluated and a plan for use over time needs to be put into place. Big Data environments make it possible to quickly ingest lots of data, but to use that data and manage it over time, it is still necessary to know basic facts:

- Its origin
- Its format
- What the data elements represent
- How it connects to other data
- How frequently it will be updated

As more data becomes available (like US Census Bureau Statistics, shopping demographics, weather satellite data, research data sets), data needs to be evaluated for worth and reliability. Review the available data sources, and the processes that create those sources and manage the plan for new sources.

- **Foundational data:** Consider foundational data components such as POS (Point of Sale) in a sales analysis.
- **Granularity:** Ideally, obtain data in its most granular form (not aggregated). That way it can be aggregated for a range of purposes.
- **Consistency:** If possible, select data that will appear appropriately and consistently across visualizations, or recognize limitations.
- **Reliability:** Choose data sources that are meaningful and credible over time. Use trusted, authoritative sources.
- **Inspect/profile new sources:** Test changes before adding new data sets. Unexpected material or significant changes in visualization outcomes can occur with the inclusion of new data sources.

Risks associated with data sources include privacy concerns. The ability to rapidly ingest and integrate data from a variety of sources at scale affords communities the ability to recombine data sets that were otherwise secured. Similarly, the published analysis may describe, through summary, aggregate, or modeled state, a subset of the public that make it suddenly identifiable; this is a side effect of the ability to perform mass computation on very large populations, but publish to a very specific local or region. For example, when demographics computed at a national or country level quickly become non-identifiable, but not when published after filtering for a postal code or household level.⁹²

Criteria used to select or filter data also pose a risk. These criteria should be objectively managed to avoid biases or skews. Filtering can have a material impact on visualization. Discretion is necessary when removing outliers, restricting data sets to a limited domain, or dropping sparse elements. It is common practice to focus the provisioned data to emphasize isolation results, but it must be done objectively and uniformly.⁹³ (See Chapter 2.)

2.3 Acquire and Ingest Data Sources

Once the sources are identified, they need to be found, sometimes purchased, and ingested (loaded) into the Big Data environment. During this process, capture critical Metadata about the source, such as its origin, size, currency, and additional knowledge about content. Many ingestion engines profile data as it is ingested, providing analysts with at least partial Metadata. Once the data is in a data lake, it can be assessed for suitability for multiple analysis efforts. Because building Data Science models is an iterative process, so is data ingestion. Iteratively identify gaps in the current data asset base and onboard those sources. Explore these data sources using profiling, visualization, mining, or other Data Science methods to define model algorithm inputs, or model hypotheses.

Before integrating the data, assess its quality. Assessment can be as simple querying to find out how many fields contain null values, or as complex as running a data quality toolset or data analytic utility against the data to profile, classify, and identify relationships between data elements. Such assessment provides insight into whether the data provides a valid sample from which to work, and, if so, how the data can be stored and accessed (scattered across logical processing units [MPP], federated, distributed by key, etc.). This work involves SMEs (usually the data scientists themselves) and platform engineers.

The assessment process provides valuable insight into how the data can be integrated with other data sets, such as Master Data or historical warehouse data. It also provides information that can be used in model training sets and validation activities.

⁹² See Martin Fowler, *Datensparsamkeit*. Blog, 12 December 2013. Fowler brings into question the assumption that we should always capture as much data as possible. He points out that the “capture it all” approach brings up privacy risks. In its place, he puts forth the idea of data minimization or data sparsity (from the German term *Datensparsamkeit*) <http://bit.ly/1f9Nq8K>.

⁹³ For more information on the impact of bias, which can profoundly affect the interpretation of scientific results, consult the following websites: INFORMS is the leading international association for Operations Research and Analytics professionals. <http://bit.ly/2sANQRW>, Statistical Society of Canada: <http://bit.ly/2oz2o5H> and American Statistical Association: <http://bit.ly/1rjAmHX>.

2.4 Develop Data Hypotheses and Methods

Data Science is about building answer sets that can find meaning or insights within the data. The development of Data Science solutions entails building statistical models that find correlations and trends within and between data elements and data sets. There will be multiple answers to a question based upon inputs to a model. For example, one must choose a rate of return to calculate the future value of a financial portfolio. Models often have more than one variable so the best practice is to find deterministic outcomes – or in other words, use best guesses as to the values to be expected. However, best guesses themselves should be educated. Each model will operate depending on the analysis method chosen. It should be tested for a range of outcomes, even the ones that appear least probable.

Models depend on both the quality of input data and the soundness of the model itself. Data models can often give insight into how to correlate the information found. An example of this is using K-Means clustering to determine the number of groupings of data to analyze further. (See Chapter 13.)

2.5 Integrate / Align Data for Analysis

Preparing the data for analysis involves understanding what is in the data, finding links between data from the various sources, and aligning common data for use.

In many cases, joining data sources is more an art than a science. For example, consider one data set based upon daily updates and another based upon monthly updates. The daily data, in order to be aligned, would have to be aggregated so that there would be an alignment pattern that could be used in the Data Science investigation.

One method is to use a common model that integrates the data using a common key. Another way is to scan and join data using indexes within the database engines for similarity and record linkage algorithms and methods. Often data is inspected during the initial phases to understand how the data could be analyzed. Clustering helps determine the grouping of the data outputs. Other methods can find correlations that will be used to build the model to display results. Consider using techniques during the initial phases that will aid in understanding how the model will show results once published.

Most solutions require the integration of Master Data and Reference Data in order to interpret results of the analytics. (See Chapter 10.)

2.6 Explore Data Using Models

2.6.1 Populate Predictive Model

Configuring predictive models includes pre-populating the model with historical information concerning the customer, market, products, or other factors that are included in the model other than the triggering factor. Pre-population calculations are usually performed in advance to enable the fastest response to triggering events. For

example, customer purchase history would be needed to pre-populate a retail market-basket recommendation model. In predicting behavior of retail markets, historical price and price change information are combined with customer, demographic, and weather information.

2.6.2 Train the Model

Execute the model against the data in order to ‘train’ the model. Training includes repeated runs of the model against the data to verify assumptions. Training will result in changes to the model. Training requires balance. Avoid over-fitting by training against a limited data fold.

Model validation must be complete before transitioning to production. Address any population imbalances or data biases with model offsets that are trained and validated; this can be tweaked in production as the initial offset is gradually adjusted through actual population interactions. Optimizing feature mix can be accomplished with Bayesian co-selection, classifier inversion, or rule induction. Models can also be combined for ensemble learning where the predictor model is built by combining the collected strengths of simpler models.

Identifying outliers or anomalies (data objects that do not comply with the general behavior exhibited by the studied elements) is critical to the evaluating the model. For more volatile datasets, apply a variance test based on the average and standard deviation. Both tests can be readily applied on profiled results. It may be that the outliers are the target of the exercise, as opposed to finding and validating trends in the majority of the data.

For predictive analytics, use a real-time data stream to finish the population of the predictive model and trigger a response, which might be an alert or an event. The data stream may require special focus on design and development of an extreme low latency processing capability. In some models, the difference in value of the predictions between fractions of a second is extreme and solutions may require innovative technology with speed of light limitations.

Models can use many statistical functions and techniques that are available in open source libraries, one of which is ‘R.’ The R Project for Statistical Computing is a free software environment for statistical computing; it contains many functions as service calls.⁹⁴ Custom functions can be developed leveraging the scripting language and shared across tools, platforms, and organizations.

Once the solution design has been created and development and operation estimated, the organization may decide whether to develop the solution to predict behavior. Real-time operational analytics solutions frequently require substantial amounts of new architecture and development and could possibly not be cost effective.

2.6.3 Evaluate Model

Once the data is placed onto a platform and ready for analysis, the Data Science begins. The model is constructed, evaluated against training sets, and validated. Refinements to the business requirements are

⁹⁴ For more information, visit the R-Project website: <http://bit.ly/19WExR5>.

expected at this point and early feasibility metrics can guide the management efforts towards further processing or discarding. It is entirely possible that testing a new hypothesis will require additional data sets.

Data scientists run queries and algorithms against the data to see if any insights become apparent. Often times a number of different mathematical functions will be run to see if any insight is found (clusters in the data, patterns that start to emerge between data element periods, etc.). During this period, data scientists are often building upon insights found in iterative batches. From these, models can be developed that display the correlation between data elements and insights.

There is an ethical component to practicing Data Science and it needs to be applied when evaluating models. Models can have unexpected results or unintentionally reflect the assumptions and biases of the people who create them. Ethical training should be required for all artificial intelligence (AI) practitioners. Ideally, the curriculum for every student learning AI, computer science, or Data Science should include ethics and security topics. However, ethics alone is not sufficient. Ethics can help practitioners understand their responsibilities to all stakeholders, but ethical training needs to be augmented with the technical capability to put good intentions into practice by taking technical precautions as a system is built and tested (Executive Office, 2016). (See Chapter 2.)

2.6.4 Create Data Visualizations

Data visualization based on the model must meet the specific needs related to the purpose of the model. Each visualization should answer a question or provide an insight. Establish the purpose and parameters for the visualization: a point in time status, trends vs. exceptions, relationships between moving parts, geographical differences, or some other point.

Select the appropriate visual to fulfill that purpose. Ensure that the visualization addresses an audience; adjust the layout and complexity to highlight and simplify accordingly. Not all audiences are ready for a complex interactive chart. Support visualizations with explanatory text.

Visualizations should tell a story. Data ‘story telling’ can link new questions to the context of data exploration. Data stories must be supported by related data visualizations to have the best effect.

2.7 Deploy and Monitor

A model that meets business needs in a feasible manner can be deployed to production for ongoing monitoring. Such models will require refinement and maintenance. Several modeling techniques are available for implementation. Models can serve batch processes as well as real-time integration messages. They can also be embedded into analytics software as input into decision management systems, historical analysis, or performance management dashboards.

2.7.1 Expose Insights and Findings

The presentation of findings and data insights, usually through data visualization, is the final step in a Data Science investigation. Insights should be connected to action items so that the organization benefits from the Data Science work.

New relationships may be explored through data visualization techniques. As a model is used, changes in the underlying data and relationships may surface, telling a new story about the data.

2.7.2 Iterate with Additional Data Sources

The presentation of findings and data insights usually generates questions that start a new process of research. Data Science is iterative, so Big Data development is iterative to support it. This process of learning from a specific set of data sources often leads to the need for different or additional data sources to both support the conclusions found and to add insights to the existing model(s).

3. Tools

Advances in technology (Moore's Law, the proliferation of hand held devices, IOT, to name a few) have created the Big Data and Data Science industry. To understand the industry, one must understand its drivers. This section will explain the tools and technologies that have enabled Big Data Science to emerge.

The advent of Massively Parallel Processing (MPP) was one of the first enablers to Big Data and Data Science as it provided the means to analyze huge volumes of information in relatively short amounts of time. Finding the needle in the haystack of information, or using machinery to plow through tons of dirt to find the gold nuggets is what we are doing today. This trend will continue.

Other advances that have changed the way we look at data and information include:

- Advanced in-database analytics
- Analytics on unstructured data (Hadoop, MapReduce)
- Integration of analytic results with operational systems
- Data visualizations across multiple media and devices
- Linking structured and unstructured information using semantics
- New data sources using IOT
- Advanced visualization capabilities
- Data enrichment capabilities
- Collaboration technologies and toolsets

Existing data warehouses, data marts, and operational data stores (ODS) are being augmented to carry Big Data workload. NoSQL technologies allow storage and query of unstructured and semi-structured data.

Access to unstructured data used to occur largely through a batch query interface that resulted in slow scheduled execution and poor response times. Several NoSQL databases are now available with designs that address specific limitations in this acquisition process. Scalable distributed databases automatically provide sharding capabilities (the ability to scale across servers natively) for parallel query execution. Of course, as with any other database, structural definition and mapping to unstructured data sets remain largely manual processes.

Immediate query, reporting, and analysis capabilities can be satisfied with Big Data in-memory technologies that allow end users to construct SQL-like queries to access unstructured data. There are also adaptors to SQL for some tools that will transmit a NoSQL process and return a SQL compliant query – with limitations and caveats. Adaptor technologies can allow existing tools to be used for unstructured data query.

Decision criteria tool sets, process implementation tools, and professional services offerings can both facilitate and expedite the process of choosing an initial set of tools. As when acquiring BI tools, it is critical to evaluate all options: build, buy, or rent (provisioned as software-as-a-service). As noted in Chapter 11, cloud sourcing tools and the associated expertise should be weighed against the cost of building from scratch or deploying purchased products from vendors. Ongoing upgrade and potential replacement costs must be considered as well. Alignment to a set OLA can bridge forecasted costs and provide input into setting compelling fees and penalties for term violations.

3.1 MPP Shared-nothing Technologies and Architecture

Massively Parallel Processing (MPP) Shared-nothing Database technologies have become the standard platform for Data Science-oriented analysis of Big Data sets. In MPP databases, data is partitioned (logically distributed) across multiple processing servers (computational nodes), with each server having its own dedicated memory to process data locally. Communication between processing servers is usually controlled by a master host and occurs over a network interconnect. There is no disk sharing or memory contention, hence the name, ‘shared-nothing’.

MPP has evolved because traditional computing paradigms (indexes, distributed data sets, etc.) did not provide acceptable response times on massive tables. Even the most powerful of computing platforms (Cray computer) would take many hours or even days to compute a complex algorithm against a trillion-row table.

Consider now a number of commodity hardware servers, all lined up in a row and controlled via a host. Each is sent part of the query to run against this segmented or distributed trillion-row table. If there are, for example, 1000 processing servers, the query changes from accessing a trillion rows in one table to accessing 1000 billion-row tables. This type of computing architecture is linearly scalable, which adds to the appeal for data scientist and Big Data users requiring a scalable platform to incorporate growth.

This technology also enabled in-database analytical functions – the ability to execute analytical functions (like K-means Clustering, Regression, etc.) at the processor level. Distribution of workload to the processor level greatly speeds up analytical queries – thereby fueling innovation in Data Science.

A system that automatically distributes data and parallelizes query workloads across all available (localized) hardware is the optimum solution for Big Data analytics.

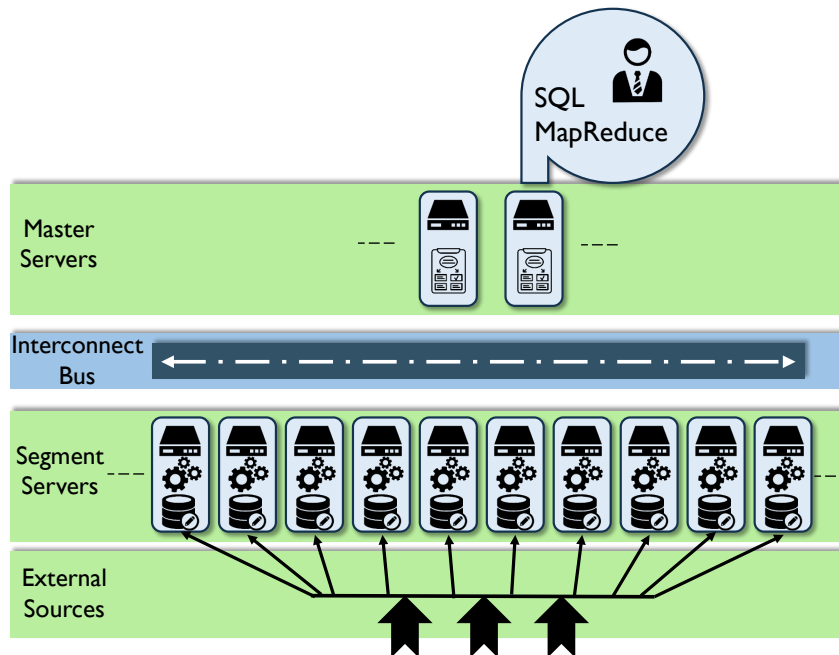


Figure 102 Columnar Appliance Architecture⁹⁵

Data volumes are growing fast. Companies can grow the capacity and performance of their systems over time by adding new nodes. MPP makes it easy to expand the parallelism of hundreds or thousands of cores across an ever-growing pool of machines. A massively parallel, shared-nothing architecture fully uses each core, with linear scalability and increased processing performance on large data sets.

3.2 Distributed File-based Databases

Distributed file-based solutions technologies, such as the open source Hadoop, are an inexpensive way to store large amounts of data in different formats. Hadoop stores files of any type – structured, semi-structured, and unstructured. Using a configuration similar to MPP Shared-nothing (an MPP foundation for file storage), it shares files across processing servers. It is ideal for storing data securely (as many copies are made), but has challenges when trying to allow access to data via structured or analytical mechanism (like SQL).

Due to its relatively low cost, Hadoop has become the landing zone of choice for many organizations. From Hadoop, data can be moved to MPP Shared-nothing databases to have algorithms run against it. Some organizations run complex Data Science queries in Hadoop, and are not concerned with response times in the order of hours and days (rather than minutes for the former architecture).

The language used in file-based solutions is called MapReduce. This language has three main steps:

- **Map:** Identify and obtain the data to be analyzed
- **Shuffle:** Combine the data according to the analytical patterns desired

⁹⁵ Image Source: “Greenplum Database 4.0: Critical Mass Innovation”, White Paper, August 2010.

- **Reduce:** Remove duplication or perform aggregation in order to reduce the size of the resulting data set to only what is required

These steps can be combined in many different tools in different ways, both in sequence and in parallel, to do complex manipulations.

3.3 In-database Algorithms

An in-database algorithm uses the principle that each of the processors in a MPP Shared-nothing platform can run queries independently, so a new form of analytics processing could be accomplished by providing mathematical and statistical functions at the computing node level. Open-source libraries of scalable in-database algorithms for machine learning, statistics, and other analytical tasks were designed both for in- and out-of-core execution, and for the shared-nothing parallelism offered by modern parallel database engines, ensuring that computation is done close to the data. By moving the computation closer to the data, the computing time is dramatically reduced for complex algorithms (such as K-means Clustering, Logistic or Linear regression, Mann-Whitney U Test, Conjugate Gradient, Cohort Analysis, etc.).

3.4 Big Data Cloud Solutions

There are vendors who provide cloud storage and integration for Big Data, including analytic capabilities. Based on defined standards, customers load their data a cloud environment. The vendor enhances the data, either as open data sets or as provided by other organizations. The customer can do analytics and Data Science using the combined data set. One application uses retail offers as the subject for the data, combines it with geographic and sales data, and offers airline miles for customers who agree to have their data used in this way.

3.5 Statistical Computing and Graphical Languages

R is an open source scripting language and environment for statistical computing and graphics. It provides a wide variety of statistical techniques such as linear and nonlinear modeling, classical statistical tests, time-series analysis, classification, and clustering. Because it is a scripting language, models developed in *R* can be implemented in a variety of environments, differing platforms and collaborated development across multiple geographic and organizational boundaries. The *R* environment can also produce publication-quality plots, including mathematical symbols and formulae, within the control of the end user.

3.6 Data Visualization Tools

Traditional tools in data visualization have both a data and a graphical component. Advanced visualization and discovery tools use in-memory architecture to allow users to interact with the data. Patterns in a large data set

can be difficult to recognize in a numbers display. A visual pattern can be picked up quickly when thousands of data points are loaded into a sophisticated display.

Information graphics or infographics are graphical representations stylized for effective interaction and comprehension. Marketing adopted these to provide visual appeal to presentations. Journalists, bloggers, and teachers found infographics useful for trend analysis, presentation, and distribution. Information visualization methods like radar charts, parallel coordinate plots, tag charts, heat maps, and data maps are now supported by many toolsets. These allow users to rapidly discern changes in data over time, gain insights into related items, and understand potential cause and effect relationships before impacts occur. These tools have several benefits over traditional visualization tools:

- Sophisticated analysis and visualization types, such as small multiples, spark lines, heat maps, histograms, waterfall charts, and bullet graphs
- Built-in adherence to visualization best practices
- Interactivity enabling visual discovery

4. Techniques

4.1 Analytic Modeling

Several open source tools are available for development, as well as cloud data processing for model development, for visual development process, for web scraping, and for linear programming optimization. To share and execute models by other applications, look for tools that support the predictive model markup language (PMML), an XML-based file format.

Real-time access can resolve many latency issues from batch processing. The Apache Mahout is an open source project aimed at creating a machine-learning library. Mahout is positioned to automate Big Data exploration through recommendation mining, document classification, and item clustering. This branch of development efforts bypasses the traditional batch query MapReduce data access techniques. Leveraging an API interface directly into the storage layer HDFS, a variety of data access techniques can be provided such as SQL, content streaming, machine learning, and graphics libraries for data visualization.

Analytic models are associated with different depths of analysis:

- **Descriptive modeling** summarizes or represents the data structures in a compact manner. This approach does not always validate a causal hypothesis or predict outcomes. However, it does use algorithms to define or refine relationships across variables in a way that could provide input to such analysis.
- **Explanatory modeling** is the application of statistical models to data for testing causal hypothesis about theoretical constructs. While it uses techniques similar to data mining and predictive analytics,

its purpose is different. It does not predict outcomes; it seeks to match model results only with existing data.

Key to predictive analytics is to learn by example through training the model. Performance of a learning method relates its predictive abilities on independent test data. Assessment guides the choice of learning and measures the quality of the chosen model. Model selection estimates performance where assessment evaluates the generalization error on new data.

Avoid over-fitting – a situation that occurs when the model is trained against non-representative datasets, is overly complex in relation to its data, or has described noise instead of the underlying relationship(s). Use additional techniques such as K-fold validation to indicate when training is no longer resulting in better generalization.

Training error consistently decrease with model complexity and can drop off to zero. Therefore, it is not a useful estimate of the test error. Randomly divide the data set into three parts to form training, testing, and validation sets. The training set is used to fit the model, the validation set is used to predict error for selection, and the test set is used for assessment of the generalization error of the final model.

Reusing the same test-set repeatedly can underestimate the true test error. Ideally, perform cross-validation by randomly dividing the data set into a set of K-folds or cross-validation groups. Perform training on all but one set of data based on strongly correlated predictor variables. Test the model on the remaining piece and determine generalization error based on all K-folds. Several statistical tests can be applied and performed to numerically assess contextual model validity.

4.2 Big Data Modeling

Modeling Big Data is a technical challenge but critical if an organization that wants to describe and govern its data. Traditional Enterprise Data Architecture principles do apply; data needs to be integrated, specified, and managed.

The main driver to physically model a data warehouse is to enable population of data for query performance. This driver is not in play for Big Data. This is not an excuse to abandon the modeling process or to hand it off to a developer. The value of modeling the data is that it enables people to understand data content. Apply proven data modeling techniques while accounting for the variety of sources. Develop the subject area model, at least in a summarized way, so it can be related to proper contextual entities and placed into the overall roadmap, just like any other kind of data. The challenge is to make an understandable and useful picture out of these large data sets, and for a justifiable cost.

Understand how the data links between data sets. For data of different granularity, prevent combinations that count data elements or values more than once; for example, don't combine atomic and aggregate sets.

5. Implementation Guidelines

Many of the general principles of managing warehouse data apply to managing Big Data: ensuring that the data sources are reliable, having sufficient Metadata to enable data use, managing the quality of data, figuring out how to integrate data from different sources, and ensuring that data is secure and protected. (See Chapters 6, 7, and 8.) The differences in implementing a Big Data environment are connected to a set of unknowns: how the data will be used, which data will be valuable, how long it needs to be retained.

Data velocity may lead people to think they do not have time to implement controls. This is a dangerous assumption. With larger data sets, managing ingestion and inventorying data in a lake is critical to preventing it from becoming a swamp.

Ingestion may not always require organizational ownership or commitment to the data set being studied. Consider leasing a Big Data platform for finite periods to explore data of interest. Exploration can quickly determine which areas show potential value. Do this before ingesting into the organizational data lake, data store, or data staging area; once landed, it can be awkward to remove.

5.1 Strategy Alignment

Any Big Data / Data Science program should be strategically aligned with organizational objectives. Establishing a Big Data strategy drives activities related to user community, data security, Metadata management, including lineage, and Data Quality Management.

The strategy should document goals, approach, and governance principles. The ability to leverage Big Data requires building organizational skills and capabilities. Use capability management to align business and IT initiatives and project a roadmap. Strategy deliverables should account for managing:

- Information lifecycle
- Metadata
- Data quality
- Data acquisition
- Data access and security
- Data governance
- Data privacy
- Learning and adoption
- Operations

5.2 Readiness Assessment / Risk Assessment

As with any development project, implementation of a Big Data or Data Science initiative should align with real business needs. Assess organizational readiness in relation to critical success factors:

- **Business relevance:** How well do the Big Data / Data Science initiatives and their corresponding use cases align with the company's business? To succeed, they must strongly enforce a business function or process.
- **Business readiness:** Is the business partner prepared for a long-term incremental delivery? Have they committed themselves to establishing centers of excellence to sustain the product in future releases? How broad is the average knowledge or skill gap within the target community and can that be crossed within a single increment?
- **Economic viability:** Has the proposed solution considered conservatively the tangible and intangible benefits? Has assessment of ownership costs accounted for the option of buying or leasing items versus building from scratch?
- **Prototype:** Can the proposed solution be prototyped for a subset of the end user community for a finite timeframe to demonstrate proposed value? Big bang implementations can cause big dollar impacts and a proving ground can mitigate these delivery risks.

Likely the most challenging decisions will be around data procurement, platform development, and resourcing.

- Many sources exist for digital data stores and not all need to be in-house owned and operated. Some can be procured while others can be leased.
- Multiple tools and techniques are on the market; matching to general needs will be a challenge.
- Securing staff with specific skills in a timely manner and retaining top talent during an implementation may require consideration of alternatives including professional services, cloud sourcing or collaborating.
- The time to build in-house talent may well exceed the delivery window.

5.3 Organization and Cultural Change

Business people must be fully engaged in order to realize benefits from the advanced analytics. A communications and education program is required to affect this. A Center of Excellence can provide training, start-up sets, design best practices, data source tips and tricks, and other point solutions or artifacts to help empower business users towards a self-service model. In addition to knowledge management, this center can provide timely communications across the developer, designer, analyst, and data consumer communities.

As with DW/BI, a Big Data implementation will bring together of a number of key cross-functional roles, including:

- **Big Data Platform Architect:** Hardware, operating systems, filesystems, and services.
- **Ingestion Architect:** Data analysis, systems of record, data modeling, and data mapping. Provides or supports mapping of sources to the Hadoop cluster for query and analysis.
- **Metadata Specialist:** Metadata interfaces, Metadata architecture, and contents.
- **Analytic Design Lead:** End user analytic design, best practice guidance implementation in related toolsets, and end user result set facilitation.

- **Data Scientist:** Provides architecture and model design consultation based on theoretical knowledge of statistics and computability, delivery on appropriate tools and technical application to functional requirements.

6. Big Data and Data Science Governance

Big Data, like other data, requires governance. Sourcing, source analysis, ingestion, enrichment, and publishing processes require business as well as technical controls, addressing such questions as:

- **Sourcing:** What to source, when to source, what is the best source of data for particular study
- **Sharing:** What data sharing agreements and contracts to enter into, terms and conditions both inside and outside the organization
- **Metadata:** What the data means on the source side, how to interpret the results on the output side
- **Enrichment:** Whether to enrich the data, how to enrich the data, and the benefits of enriching the data
- **Access:** What to publish, to whom, how, and when

An enterprise view of data should drive decisions on data handling.

6.1 Visualization Channels Management

A critical success factor in implementing a Data Science approach is the alignment of the appropriate visualization tools to the user community. Depending on the size and nature of the organization, there are likely many different visualization tools being applied in a variety of processes. Ensure that users understand the relative complexity of the visualization tools. Sophisticated users will have increasingly complex demands. Coordination between enterprise architecture, portfolio management, and maintenance teams will be necessary to control visualization channels within and across the portfolio. Be aware that changing data providers or selection criteria will likely have downstream impacts to the elements available for visualization, which can impact the effectiveness of tools.

6.2 Data Science and Visualization Standards

It is a best practice to establish a community that defines and publishes visualization standards and guidelines and reviews artifacts within a specified delivery method; this is particularly vital for customer- and regulatory-facing content. Standards may include:

- Tools standards by analytic paradigm, user community, subject area
- Requests for new data
- Data set process standard

- Processes for neutral and expert presentation to avoid biased results, and to ensure that all elements included have been done so in a fair and consistent manner including:
 - Data inclusion and exclusion
 - Assumptions in the models
 - Statistical validity of results
 - Validity of interpretation of results
 - Appropriate methods applied

6.3 Data Security

Having a reliable process to secure data is itself an organizational asset. Policies for handling and securing Big Data should be established and monitored. These policies should account for how to prevent misuse of personal data and to secure it through its overall lifecycle.

Securely provision appropriate levels of data for authorized personnel and make subscription data accessible according to agreed-upon levels. Align services to user communities so that special services can be created to provision private data for those communities allowed to ingest it, and mask the data for others. Often organizations create policies for access to information that are not to be violated (such as no access by name, address, or phone number). In order to secure information that is highly sensitive (social security number, credit card numbers, etc.), data will be stored using encryption techniques that obfuscate the information. Encryption can be chosen that, for example, has the same ‘content’ when encrypted, so that patterns may be exposed without knowing the actual values.

Recombination measures the ability to reconstitute sensitive or private data. This capability must be managed as part of the Big Data security practice. The outcomes of the analysis may violate privacy, even though the actual data elements can only be inferred. Understanding the outcomes at the Metadata Management level is critical to avoid this and other potential security violations. This requires knowledge of the intended consumption or analysis to be performed and by what role. Some trusted persons within the organization will be granted the ability to read this data when necessary, but not everyone, and certainly not for deep analysis. (See Chapters 2 and 7.)

6.4 Metadata

As part of a Big Data initiative, an organization will bring together data sets that were created using different approaches and standards. Integration of such data is challenging. Metadata related to these data sets is critical to their successful use. Metadata needs to be carefully managed as part of data ingestion, or the data lake will quickly become a data swamp. The user community must have tools that enable them to create a master list of data sets with Metadata that characterizes the structure, content, and quality of the data, including the source and lineage of the data and the definition and intended uses of entities and data elements. Technical Metadata can be harvested from a variety of Big Data tools including data storage layers, data integration, MDM, and

even the source filesystems. Consideration of real-time feeds versus data at rest versus computational data elements is necessary to complete the source side lineage.

6.5 Data Quality

Data Quality is a measure of deviation from an expected result: the smaller the difference, the better the data meets expectation, and the higher the quality. In an engineered environment, standards for quality should be easy to define (though practice shows that they are not or that many organizations do not take the time to define them). Some people have raised the question of whether data quality even matters for Big Data. Common sense says it does. For analytics to be reliable, the underlying data must be reliable. In Big Data projects, it may seem very difficult to determine the quality of data, but an effort needs to be made to assess quality in order to have confidence in the analysis. This can be done through an initial assessment, which is necessary to understand the data, and through that, the identification of measurements for subsequent instances of the data set. Data quality assessment will produce valuable Metadata that will be necessary input to any effort to integrate data.

Consider that most mature Big Data organizations scan data input sources using data quality toolsets to understand the information contained within. Most advanced data quality toolsets offer functionality that enables an organization to test assumptions and build knowledge about its data. For example:

- **Discovery:** Where information resides within the data set
- **Classification:** What types of information are present based upon standardized patterns
- **Profiling:** How the data is populated and structured
- **Mapping:** What other data sets can be matched to these values

Just as in DW/BI, it is tempting to put data quality assessment last. Without it, though, it may be difficult to know what Big Data represents or how to make connections between data sets. Integration will be necessary, and the likelihood that data feeds will be provisioned with identical structures and elements is very nearly zero. This means for example, codes and other potential linking data will likely vary from data provider to data provider. Without initial assessment, such conditions will go unnoticed until an analytic need is expressed that attempts to merge or combine those providers.

6.6 Metrics

Metrics are vital to any management process; they not only quantify activity, but can define the variation between what is observed and what is desired.

6.6.1 Technical Usage Metrics

Many of the Big Data tools offer insightful administrator reporting capabilities that interact directly with the contents queried by the user community. Technical usage analysis looks for data hot spots (most frequently

accessed data) in order to manage data distribution and preserve performance. Growth rates also feed into capacity planning.

6.6.2 Loading and Scanning Metrics

Loading and scanning metrics define the ingestion rate and interaction with the user community. As each new data source is acquired, loading metrics are expected to spike and then level as that source is fully ingested. Real-time feeds may be served through service queries, but can also appear as scheduled extracts are processed; for these feeds, expect a constant increase in data loading.

The application layer(s) would likely provide the best data usage metrics from execution logs. Monitor the consumption or access through available Metadata, which can guide usage analysis by showing query execution plans that have occurred most frequently.

Scanning metrics should be combined with any query processing that may occur outside of the analytical processing itself. Administrative tools should be able to provide this level of reporting, as well as overall service health.

6.6.3 Learnings and Stories

In order to show value, the Big Data / Data Science program must measure tangible outcomes that justify the cost of developing solutions and managing process changes. Metrics can include quantification of benefits, cost prevention or avoidance, as well as length of time between initiation and realized benefits. Common measurements include

- Counts and accuracy of models and patterns developed
- Revenue realization from identified opportunities
- Cost reduction from avoiding identified threats

Sometimes, the outcomes of the analytics tell stories that can lead to organization re-direction, re-vitalization, and new opportunity. One measurement can be a count of new projects and initiatives generated by marketing and senior executives.

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Data Management Maturity Assessment

1. Introduction

Capability Maturity Assessment (CMA) is an approach to process improvement based on a framework – a Capability Maturity Model (CMM) – that describes how characteristics of a process evolve from ad hoc to optimal. The CMA concept grew out of efforts by the United States Department of Defense to establish criteria through which to evaluate software contractors. In the mid-1980s, the Capability Maturity Model for Software was published by the Software Engineering Institute of Carnegie-Mellon University. While first applied to software development, CMMs have been developed for a range of other fields, including data management.

Maturity models are defined in terms of a progression through levels that describe process characteristics. When an organization gains an understanding of process characteristics, it can evaluate its level of maturity and put in place a plan to improve its capabilities. It can also measure improvement and compare itself to competitors or partners, guided by the levels of the model. With each new level, process execution becomes more consistent, predictable, and reliable. Processes improve as they take on characteristics of the levels. Progression happens in a set order. No level can be skipped. Levels commonly include:⁹⁶

- **Level 0:** Absence of capability
- **Level 1:** Initial or Ad Hoc: Success depends on the competence of individuals
- **Level 2:** Repeatable: Minimum process discipline is in place
- **Level 3:** Defined: Standards are set and used
- **Level 4:** Managed: Processes are quantified and controlled
- **Level 5:** Optimized: Process improvement goals are quantified

Within each level, criteria are described across process features. For example, a maturity model may include criteria related to how processes are executed, including the level of automation of those processes. It may focus on policies and controls, as well as process details.

⁹⁶ Adapted from Select Business Solutions, “What is the Capability Maturity Model?” <http://bit.ly/IFMJ18> (Accessed 2016-11-10).

Such an assessment helps identify what is working well, what is not working well, and where an organization has gaps. Based on the findings, the organization can develop a roadmap to target:

- High-value improvement opportunities related to processes, methods, resources, and automation
- Capabilities that align with business strategy
- Governance processes for periodic evaluation of organizational progress based on characteristics in the model

A Data Management Maturity Assessment (DMMA) can be used to evaluate data management overall, or it can be used to focus on a single Knowledge Area or even a single process. Whatever the focus, a DMMA can help bridge the gap between business and IT perspectives on the health and effectiveness of data management practices. A DMMA provides a common language for depicting what progress looks like across Data Management Knowledge Areas and offers a stage-based path to improvement, which can be tailored to an organization's strategic priorities.⁹⁷ Thus, it can be used both to set and to measure organizational goals, as well as to compare one's organization against other organizations or industry benchmarks.

Before beginning any DMMA, an organization has to establish a baseline understanding of its current state capabilities, assets, goals, and priorities. A certain level of organizational maturity is required to conduct the assessment in the first place, as well as to effectively respond to the assessment results by setting targets, establishing a roadmap, and monitoring progress.

1.1 Business Drivers

Organizations conduct capability maturity assessments for a number of reasons:

- **Regulation:** Regulatory oversight requires minimum levels of maturity in data management.
- **Data Governance:** The data governance function requires a maturity assessment for planning and compliance purposes.
- **Organizational readiness for process improvement:** An organization recognizes a need to improve its practices and begins by assessing its current state. For example, it makes a commitment to manage Master Data and needs to assess its readiness to deploy MDM processes and tools.
- **Organizational change:** An organizational change, such as a merger, presents data management challenges. A DMMA provides input for planning to meet these challenges.
- **New technology:** Advancements in technology offers new ways to manage and use data. The organization wants to understand the likelihood of successful adoption.
- **Data management issues:** There is need to address data quality issues or other data management challenges and the organization wants to baseline its current state in order to make better decisions about how to implement change.

⁹⁷ <http://bit.ly/1Vev9xx> July 18 2015.

Data Management Maturity Assessment

Definition: A method for ranking practices for handling data within an organization to characterize the current state of data management and its impact on the organization

Goals:

1. To comprehensively discover and evaluate critical data management activities across an organization.
2. To educate stakeholder about concepts, principles, and practices of data management, as well as to identify their roles and responsibilities in a broader context as the creators and managers of data.
3. To establish or enhance a sustainable enterprise-wide data management program in support of operational and strategic goals.

Business
Drivers



(P) Planning, (C) Control, (D) Development, (O) Operations

Figure 103 Context Diagram: Data Management Maturity Assessment

1.2 Goals and Principles

The primary goal of a data management capability assessment is to evaluate the current state of critical data management activities in order to plan for improvement. The evaluation places the organization on the maturity scale by clarifying specific strengths and weaknesses. It helps the organization identify, prioritize, and implement improvement opportunities.

In meeting its primary goal, a DMMA can have a positive impact on culture. It helps:

- Educate stakeholders about data management concepts, principles, and practices
- Clarify stakeholder roles and responsibilities in relation to organizational data
- Highlight the need to manage data as a critical asset
- Broaden recognition of data management activities across the organization
- Contribute to improving the collaboration necessary for effective data governance

Based on assessment results, an organization can enhance its Data Management program so it supports the organization's operational and strategic direction. Typically, Data Management programs develop in organizational silos. They rarely begin with an enterprise view of the data. A DMMA can equip the organization to develop a cohesive vision that supports overall organizational strategy. A DMMA enables the organization to clarify priorities, crystalize objectives, and develop an integrated plan for improvement.

1.3 Essential Concepts

1.3.1 Assessment Levels and Characteristics

CMMs usually define five or six levels of maturity, each with its own characteristics that span from non-existent or ad hoc to optimized or high performance. See Figure 104 for a sample visualization.

The following is a generic summary of macro states of data management maturity. A detailed assessment would include criteria for sub-categories like strategy, policy, standards, role definition, etc. within each of the Knowledge Areas.

- **Level 0: No Capability:** No organized data management practices or formal enterprise processes for managing data. Very few organizations exist at a Level 0. This level is acknowledged in a maturity model for purposes of definition.
- **Level 1 Initial / Ad Hoc:** General-purpose data management using a limited tool set, with little or no governance. Data handling is highly reliant on a few experts. Roles and responsibilities are defined within silos. Each data owner receives, generates, and sends data autonomously. Controls, if they exist, are applied inconsistently. Solutions for managing data are limited. Data quality issues are pervasive but not addressed. Infrastructure supports are at the business unit level.

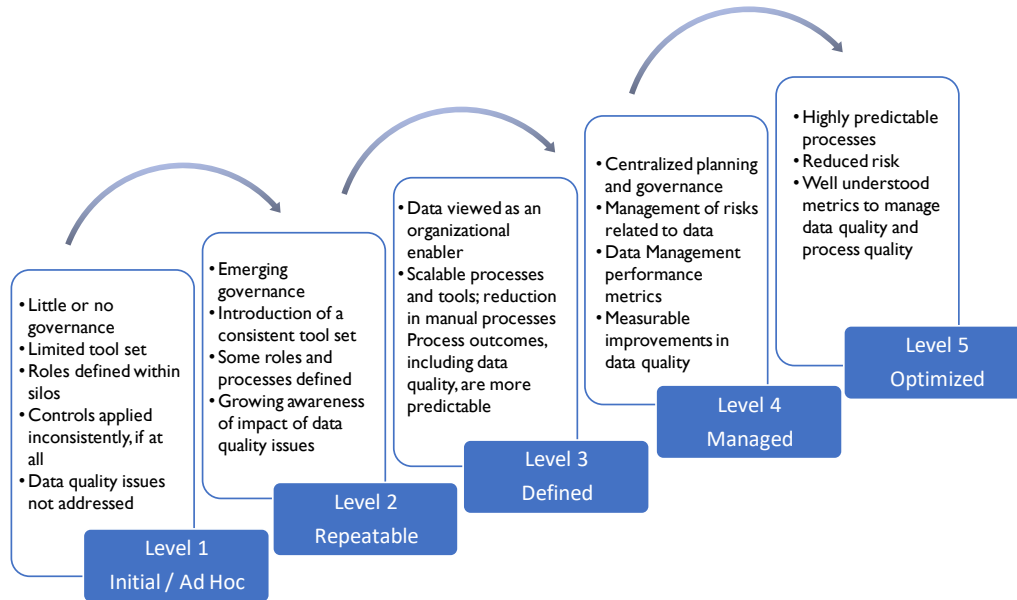


Figure 104 Data Management Maturity Model Example

Assessment criteria may include the presence of any process controls, such as logging of data quality issues.

- **Level 2 Repeatable:** Emergence of consistent tools and role definition to support process execution. In Level 2, the organization begins to use centralized tools and to provide more oversight for data management. Roles are defined and processes are not dependent solely on specific experts. There is organizational awareness of data quality issues and concepts. Concepts of Master and Reference Data begin to be recognized.

Assessment criteria might include formal role definition in artifacts like job descriptions, the existence of process documentation, and the capacity to leverage tool sets.

- **Level 3 Defined:** Emerging data management capability. Level 3 sees the introduction and institutionalization of scalable data management processes and a view of DM as an organizational enabler. Characteristics include the replication of data across an organization with some controls in place and a general increase in overall data quality, along with coordinated policy definition and management. More formal process definition leads to a significant reduction in manual intervention. This, along with a centralized design process, means that process outcomes are more predictable.

Assessment criteria might include the existence of data management policies, the use of scalable processes, and the consistency of data models and system controls.

- **Level 4 Managed:** Institutional knowledge gained from growth in Levels 1-3 enables the organization to predict results when approaching new projects and tasks and to begin to manage risks related to data. Data management includes performance metrics. Characteristics of Level 4 include standardized tools for data management from desktop to infrastructure, coupled with a well-formed centralized planning and governance function. Expressions of this level are a measurable increase in data quality and organization-wide capabilities such as end-to-end data audits.

Assessment criteria might include metrics related to project success, operational metrics for systems, and data quality metrics.

- **Level 5: Optimization:** When data management practices are optimized, they are highly predictable, due to process automation and technology change management. Organizations at this level of maturity focus on continuous improvement. At Level 5, tools enable a view data across processes. The proliferation of data is controlled to prevent needless duplication. Well-understood metrics are used to manage and measure data quality and processes.

Assessment criteria might include change management artifacts and metrics on process improvement.

1.3.2 Assessment Criteria

Each capability level will have specific assessment criteria related to the processes being evaluated. For example, if the maturity of the data modeling function is being evaluated, level 1 may ask whether a data modeling practice exists at all and how many systems it extends to; level 2 may ask whether an approach to enterprise data modeling has been defined; level 3 will ask the degree to which the approach has been implemented; level 4 will ask whether modeling standards have been effectively enforced; and level 5 will ask about processes in place to improve modeling practices. (See Chapter 5.)

At any level, assessment criteria will be evaluated along a scale, such as 1 – Not started, 2 – In process, 3 – Functional, 4 – Effective, showing progress within that level and movement toward the next level. Scores can be combined or visually displayed to enable understanding of the variance between current and desired state.

When assessing using a model that can be mapped to a DAMA-DMBOK Data Management Knowledge Area, criteria could be formulated based on the categories in the Context Diagram:

- **Activity:** To what degree is the activity or process in place? Are criteria defined for effective and efficient execution? How well defined and executed is the activity? Are best practice outputs produced?
- **Tools:** To what degree is the activity automated and supported by a common set of tools? Is tool training provided within specific roles and responsibilities? Are tools available when and where needed? Are they configured optimally to provide the most effective and efficient results? To what extent is long-term technology planning in place to accommodate future state capabilities?
- **Standards:** To what degree is the activity supported by a common set of standards? How well documented are the standards? Are standards enforced and supported by governance and change management?
- **People and resources:** To what degree is the organization staffed to carry out the activity? What specific skills, training, and knowledge are necessary to execute the activity? How well are roles and responsibilities defined?

Figure 105 illustrates one way of presenting a visual summary of findings from a DMMA. For each of the capabilities (Governance, Architecture, etc.) the outer ring of the display shows the level of capability the organization has determined it needs to compete successfully. The inner ring displays the level of capability as determined via the assessment. Areas where the distance between the two rings is largest represent the greatest risks to the organization. Such a report can help set priorities. It can also be used to measure progress over time.

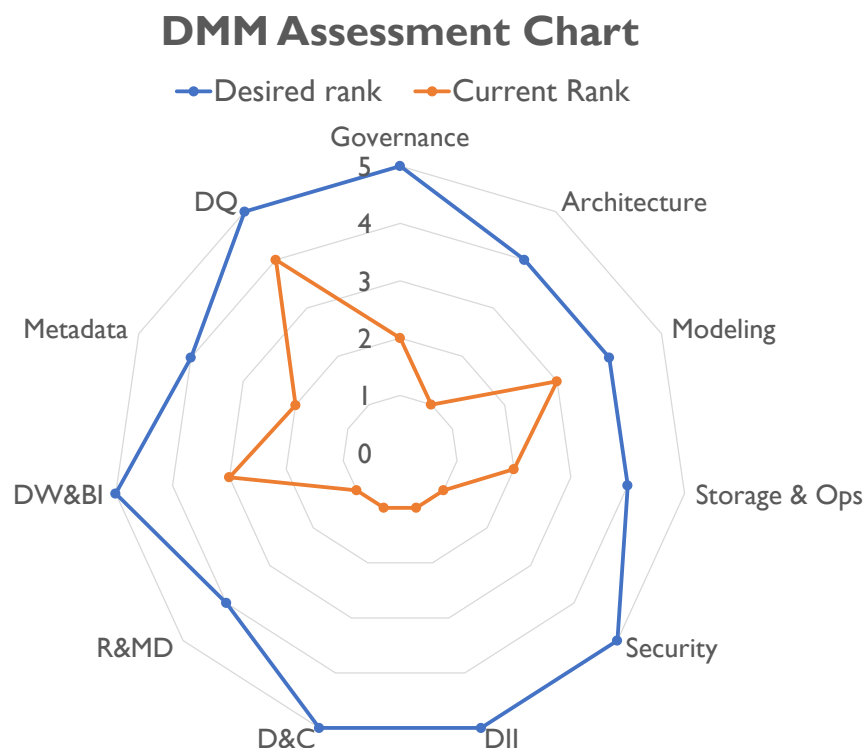


Figure 105 Example of a Data Management Maturity Assessment Visualization

1.3.3 Existing DMMA Frameworks⁹⁸

A data management maturity assessment framework is segmented into discrete data management topics. Framework focus and content vary depending on whether they have a general or industry-specific focus. However, most address subjects that can be mapped to DAMA-DMBOK Knowledge Areas. The examples below are intended to illustrate the range of Capability Maturity Models that have been developed in the data management space. Many vendors have developed their own models. Organizations should evaluate several models before choosing a vendor or before developing their own framework.

⁹⁸ For additional information and review of existing Data Management CMMs, see: Alan McSweeney, *Review of Data Management Maturity Models*, SlideShare.net, published 2013-10-23. <http://bit.ly/2spTCY9>. Jeff Gorbail, *Introduction to Data Management Maturity Models*, SlideShare.net, published 2016-08-01. McSweeney includes the DAMA-DMBOK as one of his maturity models, although the DMBOK is not structured as such.

1.3.3.1 CMMI Data Management Maturity Model (DMM)

The CMMI (Capability Maturity Model Institute) has developed the CMMI-DMM (Data Management Maturity Model) which provides assessment criteria for the following data management areas:

- Data Management Strategy
- Data Governance
- Data Quality
- Platform and Architecture
- Data Operations
- Supporting Processes

Within each of these processes, the model identifies sub-processes for evaluation. For example, the Data Quality section accounts for Data Quality Strategy and Data Quality Assessment, Profiling, and Cleansing. The model also accounts for the relation between the data management areas. For example the need for stakeholder alignment and the relation between business processes and Data Quality Management.⁹⁹

1.3.3.2 EDM Council DCAM¹⁰⁰

The Enterprise Data Management Council, an industry advocacy organization for financial services headquartered in the United States, has developed the DCAM (Data Management Capability Assessment Model). The result of a membership-driven effort to get consensus on data management best practices, the DCAM describes 37 capabilities and 115 sub-capabilities associated with the development of a sustainable Data Management program. Scoring focuses on the level of stakeholder engagement, formality of process, and existence of artifacts that demonstrate the achievement of capabilities.

1.3.3.3 IBM Data Governance Council Maturity Model¹⁰¹

The IBM Data Governance Council Maturity Model was based on input from a council of 55 organizations. Council members collaborated to define a common set of observable and desired behaviors that organizations can use to evaluate and design their own data governance programs. The purpose of the model is to help organizations build consistency and quality control in governance through proven business technologies, collaborative methods, and best practices. The model is organized around four key categories:

- **Outcomes:** Data risk management and compliance, value creation
- **Enablers:** Organizational structure and awareness, policy, stewardship

⁹⁹ <http://bit.ly/1Vev9xx> accessed 2015-07-18.

¹⁰⁰ <http://bit.ly/2sqaSga> accessed 2015-07-18.

¹⁰¹ <https://ibm.co/2sRfBIn> (accessed 2016-12-04).

- **Core disciplines:** Data Quality Management, information lifecycle management, information security and privacy
- **Supporting Disciplines:** Data Architecture, classification and Metadata, audit information, logging and reporting

The IBM model is presented both as a Maturity Framework and as a set of assessment questions with answers constructed to indicate maturity levels.

1.3.3.4 Stanford Data Governance Maturity Model¹⁰²

The Stanford Data Governance Maturity Model was developed for use by the University; it was not intended to be an industry standard. Even still, it serves as a solid example of a model that provides guidance and a standard of measurement. The model focuses on data governance, not data management, but it nevertheless provides a basis for evaluating data management overall. The model differentiates between foundational (awareness, formalization, Metadata) and project (data stewardship, Data Quality, Master Data) components. Within each, it articulates drivers for people, policies, and capabilities. It then articulates characteristics of each level of maturity. It also provides qualitative and quantitative measurements for each level.

1.3.3.5 Gartner's Enterprise Information Management Maturity Model

Gartner has published an EIM maturity model, which establishes criteria for evaluating vision, strategy, metrics, governance, roles and responsibilities, lifecycle, and infrastructure.

2. Activities

Data Management Maturity Assessments require planning. To ensure practical, actionable results, allow time within the plan for preparation of materials and evaluation of results. Assessments should be conducted in a short, defined timeframe. The purpose of the evaluation is expose current strengths and opportunities for improvement – not to solve problems.

Evaluations are conducted by soliciting knowledge from business, data management, and information technology participants. The goal is to reach a consensus view of current state capabilities, supported by evidence. Evidence may come from examination of artifacts (such as whether database backups exist), through interviews (verifying someone is performing system of record evaluation for re-use), or both.

¹⁰² <http://stanford.io/2sBR5bZ> (accessed 2016-12-04) and <http://stanford.io/2rVPyM2> (accessed 2016-12-04).

Assessments can and should be scaled to fit the needs of the organization. However, amend with care. Models may lose rigor or traceability to original intent if shortened or edited. Keep the integrity of the model intact when customizing.

2.1 Plan Assessment Activities

Planning for an assessment includes defining the overall approach and communicating with stakeholders before and during the assessment to ensure they are engaged. The assessment itself includes collecting and evaluating inputs and communicating results, recommendations, and action plans.

2.1.1 Define Objectives

Any organization that decides it should assess its data management maturity level is already engaged in the effort to improve its practices. In most cases, such an organization will have identified the drivers for the assessment. These drivers must be clarified in the form of objectives that describe the focus and influence the scope of the assessment. The objectives for the assessment must be clearly understood by executives and the lines of business, who can help ensure alignment with the organization's strategic direction.

Assessment objectives also provide criteria by which to evaluate which assessment model to adopt, which business areas to prioritize for assessment, and who should provide direct input to the process.

2.1.2 Choose a Framework

As described in Section 1.3.3, existing frameworks focus on different aspects of data management. Review these frameworks in the context of assumptions about current state and assessment objectives in order to choose one that will inform the organization in meaningful ways. Focus areas of the assessment model can be customized based on organizational focus or scope.

The choice of framework influences how the assessment is conducted. The team working on it should have expertise in the model and the methodology on which it depends.

2.1.3 Define Organizational Scope

Most DMM Frameworks are designed to apply to an entire enterprise. However, an enterprise-wide scope may be impractical. For a first assessment, it is usually best to define a manageable scope, such as a single business area or program. The areas chosen represent a meaningful subset of the organization and participants should be able to influence key business processes that affect the data assets within scope. As part of a phased approach,

evaluation can be repeated for other parts of the organization. There are trade-offs between local and enterprise assessments:

- **Localized assessments** can go much deeper into the details. They can also be done more quickly because the scope is contained. To do a localized assessment, select a function that is highly regulated, such as financial reporting within a public company. The inputs, roles, tools, and consumers may be outside of the functions being assessed, which can complicate the scoping and execution of the assessment. Well-planned localized assessments can often be aggregated and weighted to form an enterprise assessment, since many data assets are shared.
- **Enterprise assessments** focus on the broad and sometimes disconnected parts of an organization. An enterprise assessment may be created from localized DMMA's or it can be a separate undertaking. For example, an organization may evaluate different functions (research and development, manufacturing, and financing) based on the same criteria. The inputs, roles, tools, and consumers are typically pan-enterprise and multi-leveled.

2.1.4 Define Interaction Approach

In conducting a DMMA, an organization should follow recommendations for the selected model. Information gathering activities may include workshops, interviews, surveys, and artifact reviews. Employ methods that work well within the organizational culture, minimize the time commitment from participants, and enable the assessment to be completed quickly so that actions from the assessment can be defined while the process is fresh in participants' minds.

In all cases, responses will need to be formalized by having participants rate the assessment criteria. In many cases, assessment will also include actual inspection and evaluation of artifacts and other evidence.

If there are delays in completing the assessment, stakeholders are likely to lose enthusiasm for the Data Management program and the impetus for contributing to positive change. It is advisable to avoid detailed and comprehensive analysis and to emphasize sound judgment based on the expertise of the assessment leaders. The DMM Frameworks provide the measurement criteria and an embedded path to improvement. These enable synthesis of a complete picture of the current Data Management program and its parts.

2.1.5 Plan Communications

Communications contribute to the overall success of the assessment and the action items coming out of it. Communication will be directed at participants and other stakeholders. Findings may impact people's jobs, through changes in methodology and organizational alignment, so it is important to communicate clearly about the purpose, the process, and specific expectations for individuals and groups. Ensure participants understand the assessment model, as well as how the findings will be used.

Before the assessment begins, stakeholders should be informed about expectations for the assessment. Communications should describe:

- The purpose of the DMMA
- How it will be conducted
- What their involvement may be
- The schedule of assessment activities

During any activity of the assessment (for example, a focus group meeting), ensure there is a clear agenda, including a plan to answer any follow up questions. Continually remind participants of the goals and objectives. Always thank the participants and describe next steps.

Determine if the planned approach is likely to be successful across the targeted business scope, including such factors as resistance / cooperation, possible internal legal concerns about exposure to outside inspection if troubling gaps are found, or possible Human Resources concerns.

The communications plan should include a schedule to report on findings and recommendations at all levels, including general reports and executive briefings.

2.2 Perform Maturity Assessment

2.2.1 *Gather Information*

The next step is to gather appropriate inputs for the assessment, based on the interaction model. At a minimum, the information gathered will include formal ratings of assessment criteria. It may also include input from interviews and focus groups, system analysis and design documentation, data investigation, email strings, procedure manuals, standards, policies, file repositories, approval workflows, various work products, Metadata repositories, data and integration reference architectures, templates, and forms.

2.2.2 *Perform the Assessment*

The overall rating assignments and interpretation are typically multi-phased. Participants will have different opinions generating different ratings across the assessment topics. Discussion and rationalization will be needed to reconcile the ratings. Input is provided by the participants and then refined through artifact reviews or examination by the assessment team. The goal is to come to a consensus view of current state. This view should be supported by evidence (i.e., proof of practice demonstrated by behavior and artifacts). If stakeholders do not have consensus on current state, it is difficult to have consensus on how to improve the organization.

The refinement generally works as follows:

- Review results against the rating method and assign a preliminary rating to each work product or activity.
- Document the supporting evidence.
- Review with participants to come to consensus on a final rating for each area. If appropriate, use weight modifiers based on the importance of each criterion.
- Document the interpretation of the rating using the model criteria statements and assessor comments.
- Develop visualizations to illustrate results of the assessment.

2.3 Interpret Results

Interpretation of the results consists of identifying improvement opportunities aligned with organizational strategy and recommending actions required to take advantage of these opportunities. In other words, interpretation defines next steps toward a target state. When the assessment is complete, organizations need to plan for the target state that they aspire to achieve in data management. The amount of time and effort required to achieve the desired target will vary, depending on the starting point, the culture of the organization, and the drivers for change.

When presenting assessment results, start with the meaning of the ratings for the organization. The ratings can be expressed with respect to organizational and cultural drivers as well as business goals, such as customer satisfaction or increased sales. Illustrate the linkage between the current capabilities of the organization and the business processes and strategies that they support, and the benefits of improving these capabilities by moving to the target state.

2.3.1 Report Assessment Results

The assessment report should include:

- Business drivers for the assessment
- Overall results of the assessment
- Ratings by topic with gaps indicated
- A recommended approach to close gaps
- Strengths of the organization as observed
- Risks to progress
- Investment and outcomes options
- Governance and metrics to measure progress
- Resource analysis and potential future utilization
- Artifacts that can be used or re-used within the organization

The assessment report is an input to the enhancement of the Data Management program, either as a whole or by Data Management Knowledge Area. From it, the organization can develop or advance its data management

strategy. Strategy should include initiatives that further business goals through improved governance of processes and standards.

2.3.2 Develop Executive Briefings

The assessment team should prepare executive briefings that summarize findings – strengths, gaps, and recommendations – that executives will use as input to decisions about targets, initiatives, and timelines. The team must tailor the messages to clarify likely impacts and benefits for each executive group.

Often executives wish to aim higher than the assessment recommendations. In other words, they want to skip levels in the maturity model. Targeting a higher level of maturity has to be reflected in the impact analysis for the recommendations. There is a cost to this kind of acceleration, and costs must be balanced against benefits.

2.4 Create a Targeted Program for Improvements

The DMMA should have a direct impact on data strategy and IT governance, as well as the Data Management program and strategy. Recommendations from the DMMA should be actionable. These should describe capabilities the organization requires. In doing so, an assessment can be a powerful tool for IT and business leaders to set organizational priorities and allocate resources.

2.4.1 Identify Actions and Create a Roadmap

DMMA ratings highlight items for management attention. Initially, a rating is likely to be used as a standalone metric to determine how well an organization is doing a specific activity. However, ratings can be quickly operationalized into ongoing measures, especially for activities where change is desired (e.g., “The target is level ‘n’ because we need or want to be able to do something ‘z’”). If the assessment model is used for ongoing measurement, its criteria not only guides the organization to higher levels of maturity, its criteria also keeps organizational attention on improvement efforts.

The DMM assessment results should be detailed and comprehensive enough to support a multiple year data management improvement program, including initiatives that will build data management capability as the organization adopt best practices. Since change largely happens in organizations through projects, new projects must be influenced to adopt better practices. The roadmap or reference plan should contain:

- Sequenced activities to effect improvements in specific data management functions
- A timeline for implementing improvement activities
- Expected improvements in DMMA ratings once activities have been implemented
- Oversight activities, including the maturing this oversight over the timeline

The roadmap will give targets and a pace for change within prioritized work streams, and accompanied by an approach for measuring progress.

2.5 Re-assess Maturity

Re-assessments should be conducted at regular intervals. They are part of the cycle of continuous improvement:

- Establish a baseline rating through the first assessment
- Define re-assessment parameters, including organizational scope
- Repeat DMM assessment as necessary on a published schedule
- Track trends relative to the initial baseline
- Develop recommendations based on the re-assessment findings

Re-assessment can also re-invigorate or refocus effort. Measurable progress assists in maintaining commitment and enthusiasm across the organization. Changes to regulatory frameworks, internal or external policy, or innovations that could change the approach to governance and strategies are additional reasons to re-assess periodically.

3. Tools

- **Data Management Maturity Framework:** The primary tool used in a maturity assessment is the DMM framework itself.
- **Communication Plan:** A communication plan includes an engagement model for stakeholders, the type of information to be shared, and the schedule for sharing information.
- **Collaboration Tools:** Collaboration tools allow findings from the assessment to be shared. In addition, evidence of data management practices may be found in email, completed templates, and review documents created via standard processes for collaborative design, operations, incident tracking, reviews, and approvals.
- **Knowledge Management and Metadata Repositories:** Data standards, policies, methods, agendas, minutes of meetings or decisions, and business and technical artifacts that serve as proof of practice may be managed in these repositories. In some CMMs, lack of such repositories is an indicator of lower maturity in the organization. Metadata repositories can exist in several constructs, which may not be obvious to the participants. For example, some Business Intelligence applications rely completely on Metadata to compile their views and reports, while not referring to it as a separate distinct repository.

4. Techniques

Many techniques related to executing a DMMA are defined by the methodology of the DMM framework chosen. Techniques that are more general are described here.

4.1 Selecting a DMM Framework

The following criteria should be considered when selecting a DMM framework.

- **Accessibility:** Practices are stated in non-technical terms that convey the functional essence of the activity.
- **Comprehensiveness:** The framework addresses a broad scope of data management activities and includes business engagement, not merely IT processes.
- **Extensible and flexible:** The model is structured to enable enhancement of industry-specific or additional disciplines and can be used either in whole or in part, depending on the needs of the organization.
- **Future progress path built-in:** While specific priorities differ from organization to organization, the DMM framework outlines a logical way forward within each of the functions it describes.
- **Industry-agnostic vs. industry-specific:** Some organizations will benefit from an industry-specific approach, others from a more generic framework. Any DMM framework should also adhere to data management best practices that cross verticals.
- **Level of abstraction or detail:** Practices and evaluation criteria are expressed at a sufficient level of detail to ensure that they can be related to the organization and the work it performs.
- **Non-prescriptive:** The framework describes what needs to be performed, not how it must be performed.
- **Organized by topic:** The framework places data management activities in their appropriate context, enabling each to be evaluated separately, while recognizing the dependencies.
- **Repeatable:** The framework can be consistently interpreted, supporting repeatable results to compare an organization against others in its industry and to track progress over time.
- **Supported by a neutral, independent organization:** The model should be vendor neutral in order to avoid conflicts of interest, and widely available to ensure a broad representation of best practices.
- **Technology neutral:** The focus of the model should be on practices, rather than tools.
- **Training support included:** The model is supported by comprehensive training to enable professionals to master the framework and optimize its use.

4.2 DAMA-DMBOK Framework Use

The DAMA-DMBOK can be used to prepare for or establish criteria for a DMMA. Execution owners will see a direct linkage between segmented functions (the Knowledge Areas) and the corresponding tasks (activities).

The DMBOK Knowledge Areas, activities, and deliverables (work products) can be configured to a specific DMM framework based on the areas measured, their supporting activities, relevancy, and time available. This fast, checklist approach can be used to determine areas that need deeper analysis, represent gaps, or point to hot spots for remediation.

The DMBOK offers an additional advantage as an assessment-planning tool: There is a large community of knowledge professionals using the DMBOK as a guide across multiple industries, creating a community of practice around its use.

5. Guidelines for a DMMA

5.1 Readiness Assessment / Risk Assessment

Before conducting a maturity assessment, it is helpful to identify potential risks and some risk mitigation strategies. Table 33 summarizes risks and mitigation approaches.

Table 33 Typical Risks and Mitigations for a DMMA

Risk	Mitigation
Lack of organizational buy-in	Socialize the concepts related to the assessment. Establish benefit statements before conducting the assessment. Share articles and success stories. Engage an executive sponsor to champion the effort and review the results.
Lack of DMMA expertise Lack of time or in-house expertise Lack of communication planning or standards	Use third party resources or specialists. Require knowledge transfer and training as part of the engagement.
Lack of ‘Data Speak’ in the organization; Conversations on data quickly devolve into discussions about systems	Relate the DMMA to specific business problems or scenarios. Address in the communications plan. The DMMA will educate all participants regardless of background and technical experience. Orient participants to key concepts prior to the DMMA.
Incomplete or out-of-date assets for analysis	Flag ‘as of’ or balance the rating accordingly. For example, give a -1 to everything that is over 1 year out-of-date.
Narrow focus	Reduce the investigation depth to a simple DMMA and go to other areas for a quick assessment to establish ratings for a later comparative baseline. Conduct the first DMMA as a pilot, then apply lessons learned to address a broader scope. Present in-scope focus of proposed assessment in context of DAMA-DMBOK Knowledge Areas. Illustrate what is being left out of scope and discuss the need to include.
Inaccessible staff or systems	Reduce the horizontal scope of the DMMA by focusing only on available Knowledge Areas and staff
Surprises arise such as a regulation changes	Add flexibility into the assessment work stream and focus.

5.2 Organizational and Cultural Change

Establishing or enhancing a Data Management program includes changes to processes, methods, and tools. With these changes, culture must also change. Organizational and cultural transformation begins with acknowledging that things can be better. Measurement functions typically usher in meaningful change. The DMMA locates the organization on a maturity scale and provides a roadmap for improvement. Doing so, it can point an organization forward through change. The DMMA results should be part of a larger discussion within an organization. When properly supported by effective data governance, DMMA results can coalesce differing perspectives, result in a shared vision, and accelerate an organization's progress. (See Chapter 17.)

6. Maturity Management Governance

Typically, a DMMA is part of an overall set of data governance activities, each of which has a lifecycle. The lifecycle of a DMMA consists of the initial planning and initial assessment, followed by recommendations, an action plan, and periodic re-evaluation. The lifecycle itself should be governed.

6.1 DMMA Process Oversight

Oversight for the DMMA process belongs to the Data Governance team. If formal Data Governance is not in place, then oversight defaults to the steering committee or management layer that initiated the DMMA. The process should have an executive sponsor, ideally the CDO, to ensure improvements in data management activities map directly to business objectives.

The breadth and depth of oversight depend on the DMMA's scope. Each function involved in the process has a voice in the execution, method, results, and roadmaps coming from the overall assessment. Each involved data management area and organization function will have an independent view, but also will have a common language through the DMM framework.

6.2 Metrics

In addition to being a core component of any improvement strategy, metrics are a key communications tool. Initial DMMA metrics are the ratings representing the current state of data management. These can be periodically reassessed to show improvement trends. Each organization should develop metrics tailored to its target state roadmap. Sample metrics could include:

- **DMMA ratings:** DMMA ratings present a snapshot of the organization's capability level. The ratings may be accompanied by a description, perhaps a custom weighting for the rating across an assessment or specific topic area, and a recommended target state.

- **Resource utilization rates:** Powerful examples of metrics that help express the cost of data management in the form of head count. An example of this type of metric is: “Every resource in the organization spends 10% of their time manually aggregating data.”
- **Risk exposure** or the ability to respond to risk scenarios expresses an organization’s capabilities relative to their DMMA ratings. For example, if an organization wanted to begin a new business that required a high level of automation but their current operating model is based on manual data management (Level 1), they would be at risk of not delivering.
- **Spend management** expresses how the cost of data management is allocated across an organization and identifies the impacts of this cost on sustainability and value. These metrics overlap with data governance metrics.
 - Data management sustainability
 - Achievement of initiative goals and objectives
 - Effectiveness of communication
 - Effectiveness of education and training
 - Speed of change adoption
 - Data management value
 - Contributions to business objectives
 - Reductions in risks
 - Improved efficiency in operations
- **Inputs to the DMMA** are important to manage as they speak to the completeness of coverage, level of investigation, and detail of the scope relevant for interpretation of the scoring results. Core inputs could include the following: count, coverage, availability, number of systems, data volumes, teams involved, etc.
- **Rate of Change** The rate at which an organization is improving its capability. A baseline is established through the DMMA. Periodic reassessment is used to trend improvement.

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